Final Year Project Report

BS (COMPUTER SCIENCE)

for

“Personalized Medicine For Lungs Cancer”

Submitted By

MUBASHIR AZEEM ABBASI	**(63240)**
ABDUL BASIT	**(63305)**
MUHAMMAD UMER	**(62376)**
MAQSOOD AHMED	**(63821)**

Supervisor

Dr. Lubna Aziz

Faculty of Engineering, Sciences & Technology IQRA University EDC Campus Karachi

2026

Certification of Approval

It is to certify that the Final Year Design Project of BS (COMPUTER SCIENCE) **“Personalized Medicine for Lungs Cancer”** was researched by MUBASHIR AZEEM ABBASI (Reg-63240), ABDUL BASIT (Reg-63305), MUHAMMAD UMER (Reg-62376), and MAQSOOD AHMED (Reg-63821) under the supervision of “Dr. LUBNA AZIZ” and that in her opinion; it is fully adequate, in scope and quality for the degree of Bachelors of Science in Computer Sciences.

Supervisor

Coordinator

Head of Department

(Department of Computer science)

Declaration

We hereby declare that the work presented in this Final Year Project (FYP) entitled **"Personalized Medicine for Lung Cancer: AI-Powered Drug Response Prediction and Personalized Drug Recommendation Framework"** is our original work. This project has been completed through our own research, implementation, and analysis under the supervision of **Dr. Lubna Aziz.**

We further declare that this work has not been submitted, either wholly or partially, for the award of any degree or qualification at this or any other educational institution. All sources of information used in this project have been properly acknowledged and referenced.

Mubashir Azeem Abbasi **Abdul Basit**

Muhammad Umer **Maqsood Ahmed**

Supervisor

Acknowledgement

All honor belongs to Almighty Allah, who has given us a share of His infinite knowledge and provided us with the strength and direction to complete this difficult task.

We extend our heartfelt thanks to everyone who provided support and guidance during the successful completion of this **Final Year Design Project**. We sincerely thank our Supervisor and FYDP Coordinator, **Dr. Lubna Aziz**, for her unwavering support, motivation, and insightful feedback throughout all stages of this project. Her knowledge, guidance, and helpful advice were essential in influencing the course, caliber, and successful conclusion of our efforts. We express our gratitude to **Iqra University** for facilitating a nurturing academic setting, essential resources, and the chance to engage in this project, which greatly enhanced our learning and research journey.

In conclusion, we recognize the joint efforts, collaboration, and enthusiasm of all who played a direct or indirect role in bringing this project to fruition. This work would not have been achievable without their support and encouragement.

Abstract

Lung cancer remains one of the leading causes of cancer-related mortality worldwide, and selecting the most effective treatment for individual patients remains a significant challenge due to the complex biological heterogeneity of tumors. Recent advances in precision medicine and artificial intelligence have created new opportunities for predicting drug response and supporting personalized treatment decisions. This project presents an AI-powered framework for drug response prediction and personalized drug recommendation for Non-Small Cell Lung Cancer (NSCLC).

The proposed framework integrates gene expression profiles, mutation information, and drug molecular structures to predict the sensitivity of cancer cell lines to anticancer drugs. Gene expression data consisting of 8,147 features and mutation data consisting of 16 selected mutation features were collected and preprocessed for 92 NSCLC cell lines. Drug molecules were represented as molecular graphs constructed from SMILES strings using RDKit. An End-to-End Drug Graph Neural Network (GNN) architecture was developed to learn drug representations directly from molecular graph structures while simultaneously integrating biological information from cancer cell lines. The model was trained to predict drug response values expressed as LN(IC50).

Several experimental configurations, including attention-based and mutation-enhanced variants, were evaluated and compared with traditional machine learning approaches such as Linear Regression and XGBoost. The proposed End-to-End Drug GNN demonstrated strong predictive performance and achieved a Pearson correlation coefficient of **0.927**, RMSE of **1.053**, MAE of **0.814**, and R^2 score of **0.858** on the NSCLC dataset, outperforming the baseline machine learning models. In addition to drug response prediction, the framework was extended to generate personalized drug recommendations by ranking candidate drugs according to predicted sensitivity. The results demonstrate that integrating molecular graph learning with genomic and mutation information can effectively model drug response patterns and support personalized treatment selection. The proposed framework contributes toward the development of intelligent precision oncology systems and provides a foundation for future AI-assisted clinical decision support applications.

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List of Abbreviations

Abbreviation	Full Form
AI	Artificial Intelligence
ANN	Artificial Neural Network
CCLE	Cancer Cell Line Encyclopedia
CNN	Convolutional Neural Network
DNN	Deep Neural Network
DL	Deep Learning
DNA	Deoxyribonucleic Acid
FC	Fully Connected
FYP	Final Year Project
GAT	Graph Attention Network
GCN	Graph Convolutional Network
GDSC	Genomics of Drug Sensitivity in Cancer
GIN	Graph Isomorphism Network
GNN	Graph Neural Network
GPDRP	Graph-Based Drug Response Prediction
IC50	Half Maximal Inhibitory Concentration
LN(IC50)	Natural Logarithm of Half Maximal Inhibitory Concentration
MAE	Mean Absolute Error
ML	Machine Learning
MSE	Mean Squared Error
NSCLC	Non-Small Cell Lung Cancer
PCA	Principal Component Analysis
PyG	PyTorch Geometric
RDKit	RDKit Cheminformatics Toolkit
ReLU	Rectified Linear Unit
RMSE	Root Mean Squared Error
RNA	Ribonucleic Acid
RNA-seq	RNA Sequencing
R^2	Coefficient of Determination
SMILES	Simplified Molecular Input Line Entry System
SOTA	State-of-the-Art
TCGA	The Cancer Genome Atlas
TP53	Tumor Protein p53
VCF	Variant Call Format
XGBoost	Extreme Gradient Boosting
Z-score	Standard Score Normalization

CHAPTER 1

INTRODUCTION

1.1 Background of the Study

Cancer remains one of the leading causes of death worldwide and continues to pose a major challenge to healthcare systems, researchers, and clinicians. According to global cancer statistics, millions of new cancer cases are diagnosed annually, with a significant proportion resulting in mortality despite continuous advances in diagnosis and treatment. The complexity of cancer arises from its highly heterogeneous nature, where patients diagnosed with the same cancer type often exhibit different genetic, molecular, and clinical characteristics. This heterogeneity frequently leads to substantial variations in treatment outcomes among patients receiving the same therapeutic intervention.

Lung cancer is one of the most prevalent and lethal forms of cancer worldwide. Among its various subtypes, Non-Small Cell Lung Cancer (NSCLC) accounts for approximately 85% of all lung cancer cases and is responsible for a considerable number of cancer-related deaths each year. Despite the availability of multiple therapeutic options, including chemotherapy, targeted therapy, immunotherapy, and combination treatments, selecting the most effective drug for a specific patient remains a challenging task. The effectiveness of a particular drug may vary significantly across patients due to differences in molecular profiles, gene expression patterns, and genomic mutations.

Traditional treatment selection methods primarily rely on clinical observations, physician expertise, and standardized treatment protocols. While these approaches have contributed significantly to cancer management, they often fail to account for the underlying molecular diversity among patients. Consequently, many patients receive treatments that may not produce optimal therapeutic responses, leading to unnecessary side effects, increased healthcare costs, and delayed treatment success. These limitations have motivated the emergence of precision medicine, a healthcare paradigm that seeks to personalize treatment decisions based on the molecular characteristics of individual patients.

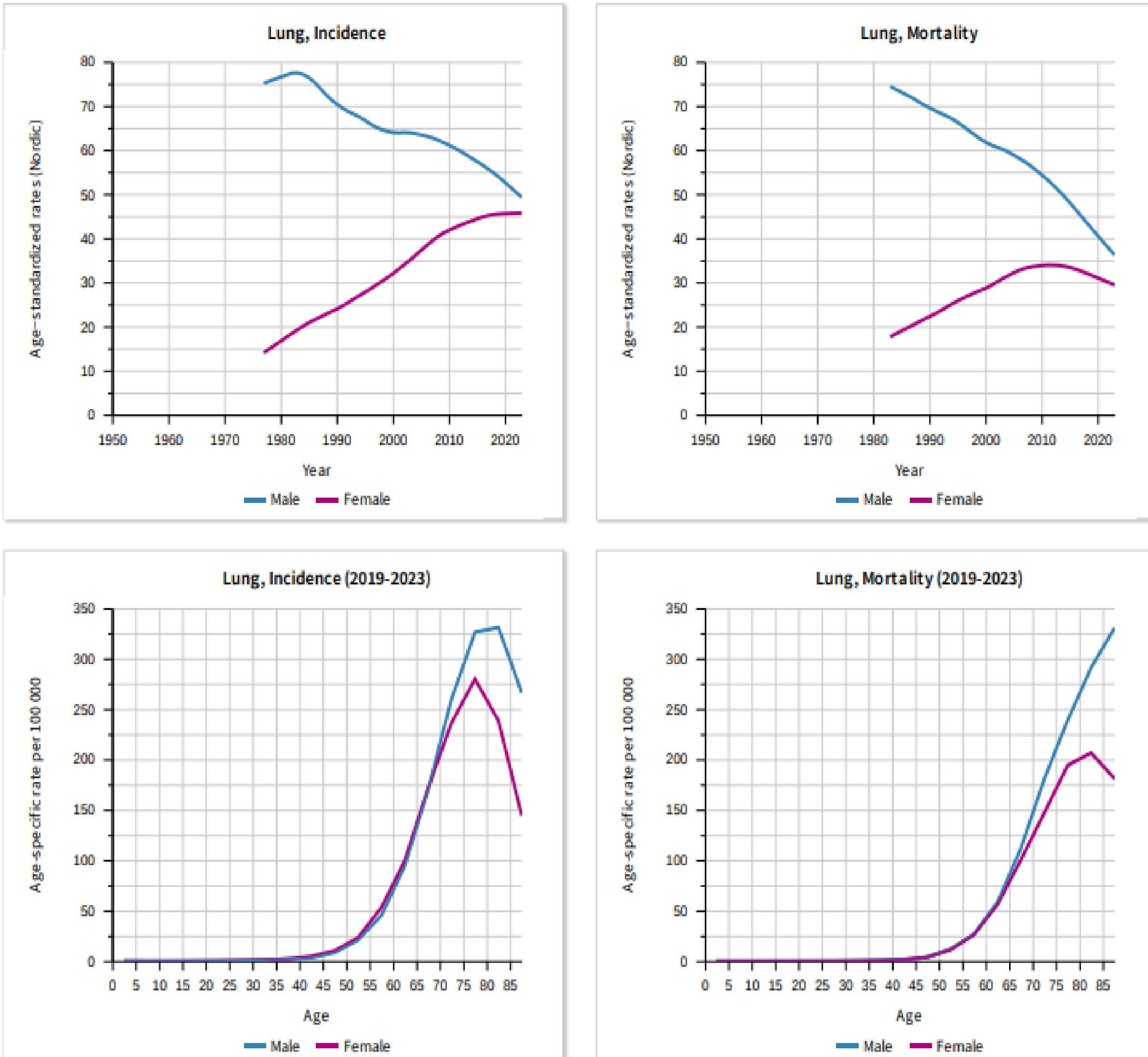

Figure 1: Global Burden of Cancer and Cancer-Related Mortality

Precision medicine aims to identify the most suitable therapeutic strategy for each patient by leveraging genomic, transcriptomic, and molecular information. In oncology, this involves analyzing biological factors such as gene expression levels, genomic mutations, molecular pathways, and drug characteristics to predict how a patient may respond to a specific treatment.

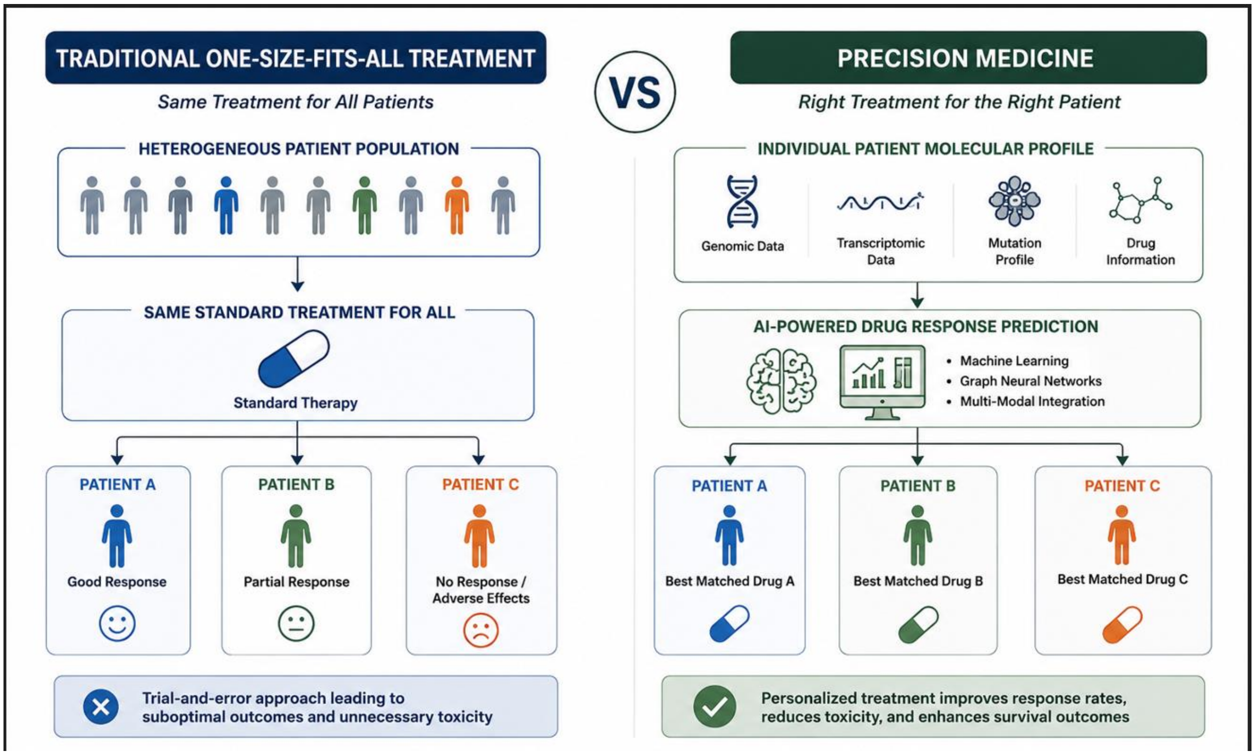

Figure 2:Traditional One-Size-Fits-All Treatment versus Precision Medicine

Advances in high-throughput sequencing technologies and large-scale pharmacogenomic initiatives have enabled researchers to collect extensive datasets containing molecular profiles and drug response measurements. These datasets provide valuable opportunities for developing computational models capable of predicting drug sensitivity and supporting personalized treatment recommendations.

Drug response prediction has emerged as a critical research area within computational biology and precision oncology. The primary objective of drug response prediction is to estimate the effectiveness of a therapeutic compound against a particular cancer cell line or patient based on molecular and pharmacological information. Accurate drug response prediction can assist clinicians in identifying effective treatments, reducing trial-and-error approaches, minimizing adverse effects, and accelerating the development of personalized therapeutic strategies.

Recent advances in Artificial Intelligence (AI), Machine Learning (ML), and Deep Learning (DL) have significantly improved the ability to model complex biological relationships. Traditional machine learning techniques such as Linear Regression, Random Forests, and Support Vector Machines have been applied to drug response prediction tasks. Although these approaches have achieved moderate success, they often struggle with the high dimensionality and non-linear nature of biological data. Deep learning methods provide a powerful alternative by automatically learning hierarchical feature representations from large and complex datasets.

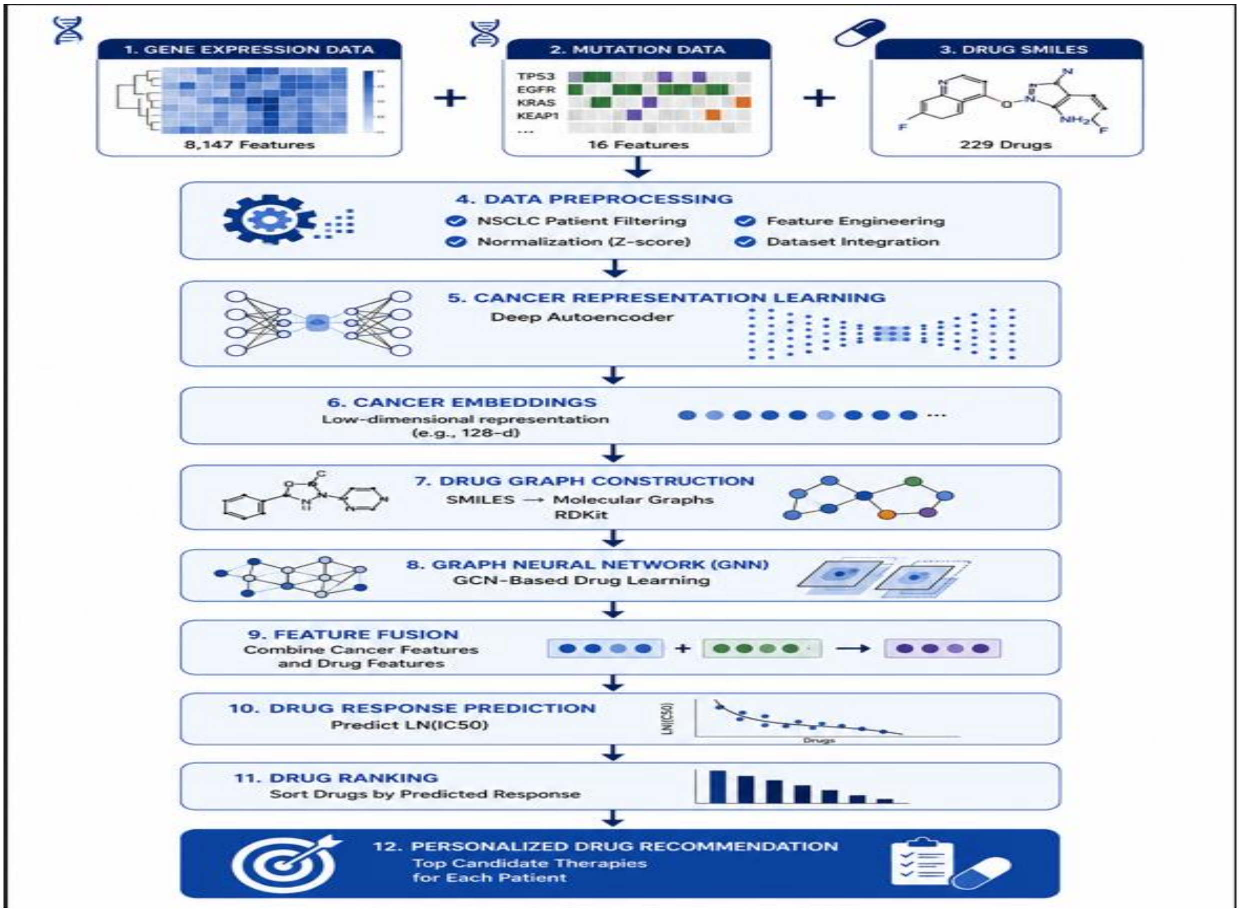

Figure 3:Drug Response Prediction Workflow

Among deep learning approaches, representation learning techniques such as Autoencoders have demonstrated considerable potential for extracting meaningful latent features from high-dimensional gene expression data. Autoencoders compress large gene expression profiles into lower-dimensional representations while preserving essential biological information. These learned representations can improve predictive performance by reducing noise and redundancy within genomic datasets. Furthermore, task-aware representation learning extends this concept by optimizing latent features not only for data reconstruction but also for the downstream prediction task, resulting in more informative cancer embeddings for drug response prediction.

In addition to cellular characteristics, the molecular structure of therapeutic compounds plays a crucial role in determining drug efficacy. Conventional drug representation methods often utilize molecular fingerprints or manually engineered descriptors. However, these approaches may fail to capture complex structural relationships within molecules. Graph Neural Networks (GNNs) address this limitation by representing molecules as graphs, where atoms correspond to nodes and chemical bonds correspond to edges. Through graph-based learning, GNNs can effectively capture structural and chemical properties of drugs, leading to improved predictive performance.

The integration of multiple biological data modalities has become increasingly important in modern drug response prediction systems. Combining gene expression profiles, mutation information, and molecular drug structures allows predictive models to capture complementary biological information and develop a more comprehensive understanding of cancer-drug interactions. Moreover, attention mechanisms have shown promise in selectively focusing on the most relevant features during multimodal fusion, thereby improving the quality of learned representations and enhancing prediction accuracy.

The present study addresses the problem of personalized drug response prediction for Non-Small Cell Lung Cancer by developing a multimodal deep learning framework that integrates task-aware cancer embeddings, mutation profiles, and graph-based drug representations. The proposed framework leverages large-scale pharmacogenomic datasets and advanced representation learning techniques to predict drug sensitivity in terms of LN(IC50) values and generate personalized drug recommendations. By combining biologically meaningful cellular features with molecular drug structures, the proposed system aims to improve prediction accuracy and contribute toward more effective precision oncology applications.

The outcomes of this research are expected to support the development of intelligent decision-support systems for cancer treatment selection, facilitate personalized therapeutic recommendations, and contribute to ongoing efforts in precision medicine and computational oncology.

1.2 Problem Statement

The successful treatment of cancer depends heavily on the ability to identify therapeutic agents that are most effective for a specific patient. However, cancer is a highly heterogeneous disease characterized by substantial genetic, molecular, and phenotypic variability among individuals. Even patients diagnosed with the same cancer subtype may exhibit significantly different responses to identical treatment regimens due to differences in gene expression patterns, genomic mutations, and molecular signaling pathways. Consequently, selecting the most appropriate drug remains one of the most challenging problems in precision oncology.

In clinical practice, treatment decisions are often based on standardized therapeutic guidelines, clinical expertise, and population-level evidence. While these approaches have improved patient outcomes, they do not fully account for the complex molecular characteristics that influence drug sensitivity and resistance. As a result, many patients receive treatments that provide limited therapeutic benefit, expose them to unnecessary toxicity, and increase healthcare costs. The inability to accurately predict individual drug responses continues to hinder the realization of truly personalized cancer treatment.

Recent advances in pharmacogenomics have enabled the collection of large-scale datasets containing gene expression profiles, mutation information, and experimentally measured drug response values. Resources such as the Genomics of Drug Sensitivity in Cancer (GDSC) and Cancer Cell Line Encyclopedia (CCLE) provide valuable opportunities for developing computational models capable of predicting drug efficacy using molecular data. However, effectively utilizing these datasets presents several challenges.

The first challenge is the high dimensionality of gene expression data. Modern transcriptomic datasets may contain thousands of genes, many of which are redundant, noisy, or only weakly associated with drug response. Traditional machine learning algorithms often struggle to learn meaningful patterns from such high-dimensional data, leading to overfitting and reduced generalization performance.

The second challenge is the limited utilization of biologically relevant mutation information. Several studies focus primarily on gene expression features while neglecting important genomic alterations that directly influence cancer progression and therapeutic response. Mutations in genes such as EGFR, KRAS, TP53, ALK, ROS1, and MET are known to significantly impact treatment outcomes in Non-Small Cell Lung Cancer. Ignoring these molecular characteristics may limit the predictive capability of drug response models.

The third challenge involves drug representation. Many existing approaches rely on handcrafted molecular descriptors or fixed fingerprints to represent therapeutic compounds. While these methods provide useful information, they often fail to capture the complex structural relationships present within molecular graphs. Consequently, valuable chemical information relevant to drug activity may be lost during feature extraction.

Another important challenge is the effective integration of heterogeneous biological data modalities. Drug response is influenced by interactions between cellular characteristics and molecular drug properties. Modeling these interactions requires frameworks capable of learning meaningful representations from multiple data sources while preserving the biological relationships between them. Simple feature concatenation approaches may not adequately capture these complex interactions.

Furthermore, many existing drug response prediction systems prioritize predictive performance while providing limited support for personalized treatment recommendation. Although predicting drug sensitivity values is important, the ultimate goal in precision oncology is to identify and rank the most effective therapeutic options for a specific patient or cancer profile. Therefore, predictive models should not only estimate drug response but also facilitate practical treatment recommendation.

To address these challenges, this study proposes a multimodal deep learning framework for personalized drug response prediction in Non-Small Cell Lung Cancer. The framework integrates task-aware cancer representations derived from gene expression data, mutation profiles associated with clinically relevant NSCLC genes, and graph-based molecular representations learned directly from drug structures. By combining these complementary sources of information within a unified predictive architecture, the proposed system aims to improve drug response prediction accuracy and generate personalized drug recommendations based on predicted LN(IC50) values.

The central problem addressed by this research can therefore be stated as follows:

How can gene expression data, mutation information, and graph-based drug representations be effectively integrated within a deep learning framework to improve drug response prediction and personalized drug recommendation for Non-Small Cell Lung Cancer?

1.3 Research Aim

The aim of this research is to develop a multimodal deep learning framework for personalized drug response prediction in Non-Small Cell Lung Cancer by integrating task-aware cancer embeddings, mutation profiles, and graph-based drug representations to accurately predict LN(IC50) values and generate effective drug recommendations.

1.4 Research Objectives

The primary objectives of this research are:

Objective 1

To preprocess and integrate pharmacogenomic datasets containing gene expression profiles, mutation information, drug response measurements, and molecular drug structures relevant to Non-Small Cell Lung Cancer.

Objective 2

To develop a task-aware representation learning framework capable of generating biologically meaningful cancer embeddings from high-dimensional gene expression data.

Objective 3

To incorporate clinically significant mutation features associated with NSCLC into the predictive framework to enhance cellular characterization.

Objective 4

To construct graph-based molecular representations of therapeutic compounds using Graph Neural Networks and learn informative drug embeddings directly from molecular structures.

Objective 5

To design a multimodal drug response prediction model that effectively combines cellular and drug representations for LN(IC50) prediction.

Objective 6

To evaluate the predictive performance of the proposed framework using standard regression metrics including Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), Coefficient of Determination (R^2), and Pearson Correlation Coefficient.

Objective 7

To develop a personalized drug recommendation mechanism capable of ranking candidate drugs according to predicted therapeutic effectiveness.

Objective	**Expected Outcome**
O1	Integrated NSCLC pharmacogenomic dataset
O2	Task-aware cancer embeddings
O3	Mutation-enhanced cellular features
O4	Learned graph-based drug representations
O5	Multimodal prediction framework
O6	Quantitative performance evaluation
O7	Personalized drug recommendation system

Table 1:Research Objectives and Expected Outcomes

1.5 Research Questions

This study seeks to answer the following research questions:

RQ1

Can task-aware representation learning generate more informative cancer embeddings for drug response prediction than conventional feature representations?

RQ2

Does the integration of mutation information improve predictive performance in NSCLC drug response prediction?

RQ3

Can Graph Neural Networks effectively learn molecular drug representations that capture structural information relevant to drug sensitivity?

RQ4

How effectively can multimodal cellular and drug information be integrated to predict LN(IC50) values for cancer-drug pairs?

RQ5

Can the proposed framework generate personalized drug recommendations that support precision oncology applications?

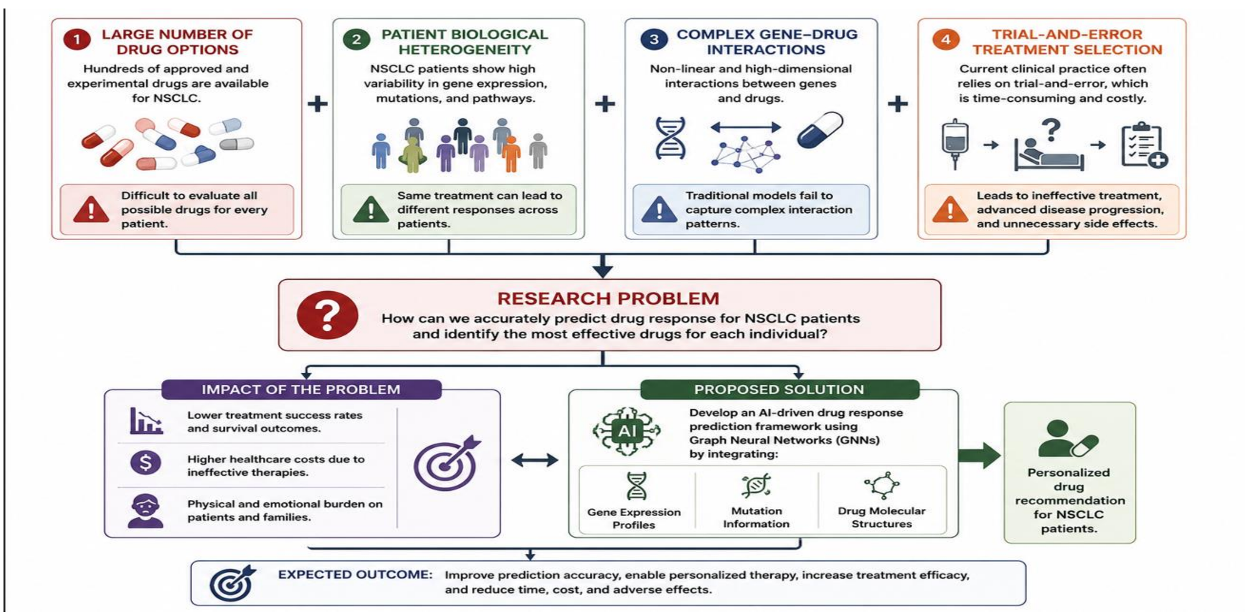

Figure 4:Research Problem Overview

1.6 Scope of the Study

The scope of this research is focused on the development and evaluation of a deep learning-based framework for personalized drug response prediction in Non-Small Cell Lung Cancer (NSCLC). The study utilizes publicly available pharmacogenomic datasets containing molecular profiles of cancer cell lines and experimentally measured drug response values. By integrating multiple biological data modalities, the proposed framework aims to improve the prediction of drug sensitivity and support personalized treatment recommendation.

The research primarily focuses on NSCLC cell lines obtained through large-scale pharmacogenomic resources. Gene expression profiles are used to capture transcriptomic characteristics of cancer cells, while mutation information from clinically significant NSCLC-related genes is incorporated to enhance biological representation. Drug information is obtained through molecular structure data represented using Simplified Molecular Input Line Entry System (SMILES) strings and transformed into graph-based representations suitable for Graph Neural Network learning.

The proposed framework employs task-aware representation learning to generate compact cancer embeddings from high-dimensional gene expression data. These embeddings are combined with mutation features and graph-based drug representations within a multimodal predictive architecture. The model predicts drug response in terms of LN(IC50) values and ranks candidate drugs according to predicted effectiveness.

The scope of this study includes:

- Non-Small Cell Lung Cancer (NSCLC) cell lines only.
- Gene expression-based cancer representation learning.
- Integration of mutation information from selected NSCLC-related genes.
- Graph-based molecular drug representation learning.
- Drug response prediction using LN(IC50) values.
- Personalized drug recommendation based on predicted drug sensitivity.
- Evaluation using standard machine learning performance metrics.

Study Boundaries

While the proposed framework provides a comprehensive approach to drug response prediction, several aspects fall outside the scope of this research.

The study does not involve clinical patient data and relies exclusively on experimentally validated cancer cell line datasets. Furthermore, the research focuses on gene expression and mutation information and does not incorporate additional omics modalities such as proteomics, metabolomics, copy number variation, methylation profiles, or histopathological imaging data.

Similarly, the proposed framework evaluates computational prediction performance and does not include laboratory validation, clinical trials, or biological experiments. Drug recommendations generated by the system are intended for research purposes and should not be interpreted as clinical treatment recommendations without further biological and clinical validation.

Included in Scope	**Outside Scope**
NSCLC Cell Lines	Clinical Patients
Gene Expression Data	Proteomics Data
Mutation Information	Metabolomics Data
Drug Graph Learning	Clinical Trials
LN(IC50) Prediction	Wet-Lab Validation
Drug Recommendation	Real-Time Hospital Deployment

Table 2:Scope and Limitations of the Study

1.7 Significance of the Study

Cancer treatment selection remains one of the most challenging problems in modern healthcare due to the complex biological variability observed among patients. Precision oncology seeks to address this challenge by utilizing molecular information to guide treatment decisions. Accurate drug response prediction systems can significantly contribute to this objective by identifying potentially effective therapies before treatment administration.

The significance of this study lies in its contribution to the growing field of AI-driven precision medicine. By integrating task-aware cancer representations, mutation profiles, and graph-based drug representations, the proposed framework aims to improve the ability of computational models to capture biologically meaningful relationships influencing drug response.

From a clinical perspective, improved drug response prediction has the potential to reduce ineffective treatments, minimize adverse drug reactions, and support personalized therapeutic planning. Although the proposed system is evaluated using cell-line data rather than patient data, it provides an important step toward developing intelligent decision-support systems for precision oncology.

For researchers in computational biology and bioinformatics, this study demonstrates the effectiveness of combining multiple representation learning techniques within a unified framework. The integration of task-aware embeddings and graph neural networks highlights the potential benefits of multimodal learning for complex biomedical prediction tasks.

The study also contributes to ongoing research in machine learning by investigating the application of deep representation learning, graph neural networks, and multimodal fusion techniques within the context of pharmacogenomics. The experimental findings may provide insights into designing future drug response prediction systems capable of leveraging diverse biological data sources.

Furthermore, the personalized drug recommendation component extends the practical utility of the proposed framework beyond simple regression prediction. By ranking candidate drugs according to predicted effectiveness, the system demonstrates how predictive models can be transformed into actionable decision-support tools for precision medicine applications.

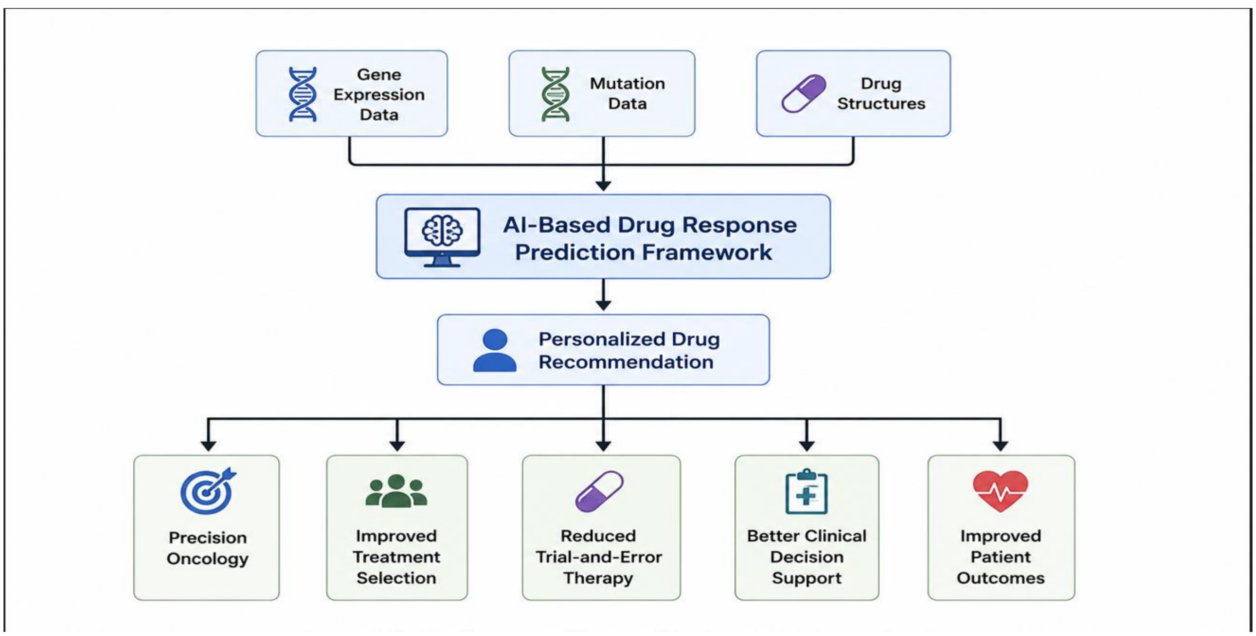

Figure 5:Significance and Impact of the Proposed Framework

1.8 Research Contributions

This research makes several important contributions to the field of computational drug response prediction and precision oncology.

Contribution 1: Development of a Task-Aware Cancer Representation Framework

A task-aware autoencoder-based representation learning framework was developed to generate compact and informative cancer embeddings from high-dimensional gene expression data. Unlike conventional autoencoders that focus solely on reconstruction, the proposed approach incorporates predictive objectives to learn representations that are more relevant to drug response prediction.

Contribution 2: Integration of Mutation Information

The study incorporates mutation information from clinically significant NSCLC-related genes into the predictive framework. This integration enables the model to capture important genomic alterations associated with drug sensitivity and resistance mechanisms.

Contribution 3: Graph-Based Drug Representation Learning

Drug molecules are represented as molecular graphs and processed using Graph Neural Networks to learn structural drug embeddings directly from molecular architecture. This approach allows the model to capture complex chemical relationships that may not be adequately represented using handcrafted molecular descriptors.

Contribution 4: Multimodal Drug Response Prediction Framework

A multimodal deep learning architecture was developed to combine cancer embeddings, mutation features, and graph-based drug representations within a unified prediction framework. This design enables the model to leverage complementary biological information from multiple sources.

Contribution 5: Comprehensive Experimental Evaluation

The proposed framework was evaluated using publicly available pharmacogenomic datasets and compared against baseline models and alternative architectures. Experimental results demonstrate substantial improvements in predictive performance across multiple evaluation metrics.

Contribution 6: Personalized Drug Recommendation System

Beyond drug response prediction, the framework was extended to generate personalized drug recommendations by ranking therapeutic compounds according to predicted LN(IC50) values. This component highlights the practical applicability of the proposed system for precision medicine research.

Contribution	Description
C1	Task-Aware Cancer Embeddings
C2	Mutation Integration
C3	Graph-Based Drug Learning
C4	Multimodal Prediction Framework
C5	Experimental Evaluation
C6	Drug Recommendation System

Table 3:Summary of Research Contributions

Chapter Summary

This chapter introduced the research background, highlighted the challenges associated with personalized drug response prediction in Non-Small Cell Lung Cancer, and established the motivation for developing advanced computational approaches. The problem statement, research aim, objectives, research questions, scope, significance, and contributions of the study were presented. Finally, the overall organization of the report was outlined. The next chapter provides a detailed review of existing literature relevant to drug response prediction, precision oncology, deep learning, and graph-based representation learning.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

The rapid advancement of precision medicine has transformed the landscape of cancer treatment by shifting therapeutic strategies from generalized protocols toward personalized interventions. Traditional treatment approaches often assume that patients diagnosed with the same cancer subtype will respond similarly to a given therapy. However, growing evidence has demonstrated that substantial molecular and genetic heterogeneity exists among cancer patients, resulting in significant differences in drug sensitivity and treatment outcomes. Consequently, identifying the most effective therapeutic agent for an individual patient remains a major challenge in modern oncology.

Drug response prediction has emerged as an important research area within computational biology, bioinformatics, and precision medicine. The primary objective of drug response prediction is to estimate the effectiveness of a therapeutic compound against a particular cancer profile using molecular and pharmacological information. Accurate prediction models can assist researchers and clinicians in identifying effective treatment options, reducing adverse drug reactions, lowering healthcare costs, and accelerating the development of personalized therapeutic strategies.

Recent advances in machine learning and deep learning have significantly improved the ability to model complex biological systems. Large-scale pharmacogenomic datasets containing gene expression profiles, mutation information, molecular drug structures, and experimentally measured drug responses have enabled the development of increasingly sophisticated predictive frameworks. Researchers have explored a variety of computational approaches ranging from traditional machine learning algorithms to advanced deep neural networks and graph-based learning methods.

Among these developments, representation learning techniques such as autoencoders have demonstrated the ability to extract meaningful latent features from high-dimensional biological data. Similarly, Graph Neural Networks (GNNs) have emerged as powerful tools for learning molecular drug representations directly from chemical structures. More recently, multimodal learning approaches have attempted to combine cellular and drug information within unified predictive frameworks capable of capturing complex drug-cell interactions.

This chapter reviews the existing literature relevant to precision oncology, pharmacogenomics, drug response prediction, deep learning, and graph-based molecular representation learning. Key state-of-the-art models are critically analyzed, their strengths and limitations are discussed, and research gaps motivating the proposed framework are identified.

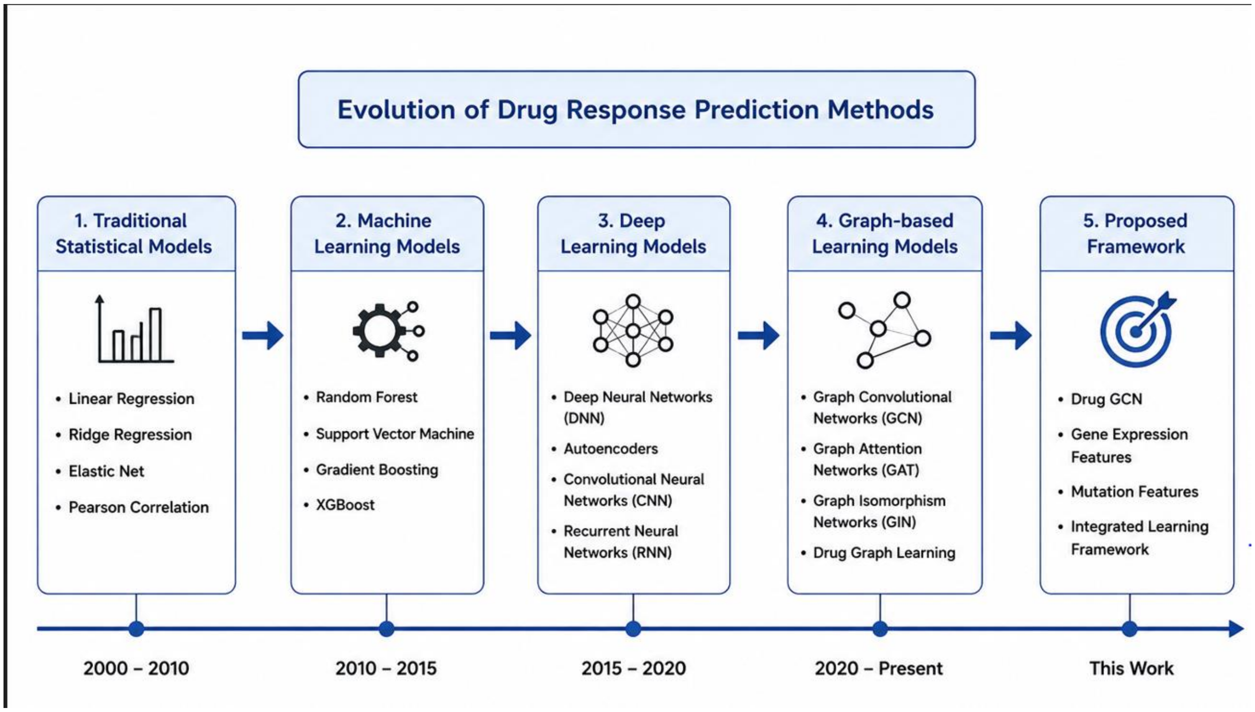

Figure 6:Evolution of Drug Response Prediction Methods

2.2 Precision Oncology and Drug Response Prediction

Cancer is a highly heterogeneous disease characterized by extensive genetic, molecular, and phenotypic variability. Even among patients diagnosed with the same cancer subtype, substantial differences may exist in gene expression patterns, genomic mutations, signaling pathways, and treatment responses. This biological heterogeneity presents a significant challenge for clinicians attempting to identify the most effective therapeutic strategy for individual patients.

Precision oncology seeks to address this challenge by tailoring treatment decisions according to the molecular characteristics of each patient. Rather than relying solely on population-level treatment guidelines, precision medicine incorporates genomic and molecular information to predict treatment outcomes and optimize therapeutic selection. The central premise of precision oncology is that patients with distinct molecular profiles may respond differently to the same drug, thereby requiring personalized treatment strategies.

Drug response prediction represents a critical component of precision oncology. The goal of drug response prediction is to estimate the effectiveness of a therapeutic compound against a specific cancer profile before treatment administration. Such predictions enable the identification of potentially effective drugs while reducing the likelihood of ineffective therapies and unnecessary toxicity.

Large-scale pharmacogenomic initiatives have significantly accelerated research in this domain by generating extensive datasets that combine molecular characteristics of cancer cell lines with experimentally measured drug responses. These resources provide valuable opportunities for developing computational models capable of learning complex relationships between cellular features and therapeutic outcomes.

One of the most widely used measurements in drug response prediction research is the half-maximal inhibitory concentration (IC50). IC50 represents the concentration of a drug required to inhibit a biological process by 50% and serves as an indicator of drug sensitivity. Lower IC50 values generally indicate greater drug effectiveness, whereas higher values suggest reduced sensitivity or drug resistance.

To improve numerical stability and reduce distributional skewness, many studies employ the natural logarithm of IC50 values, commonly referred to as LN(IC50). Predicting LN(IC50) has therefore become a standard objective in modern drug response prediction research.

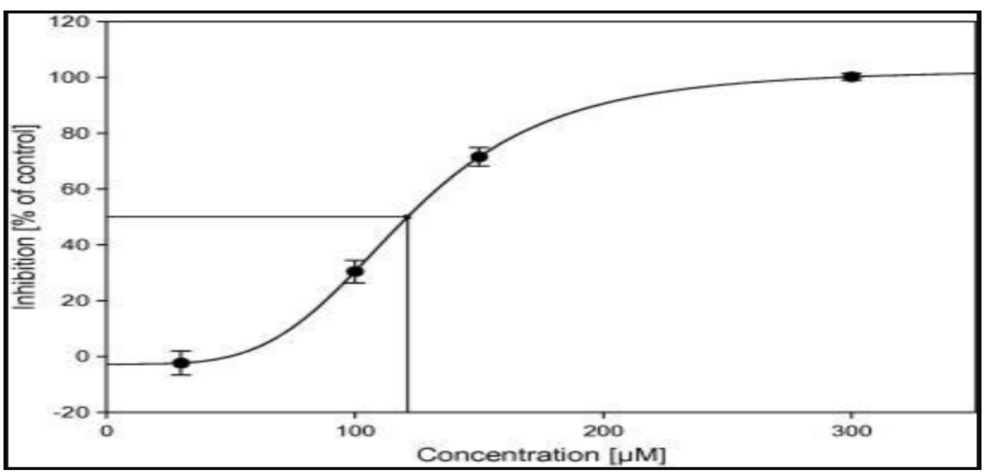

Figure 7:Drug Dose–Response Curve and IC50 Concept

The increasing availability of pharmacogenomic data, combined with advances in artificial intelligence and deep learning, has created new opportunities for developing accurate and scalable drug response prediction systems. Nevertheless, challenges associated with high-dimensional biological data, heterogeneous information sources, and complex drug-cell interactions continue to motivate ongoing research in this field.

2.3 Pharmacogenomic Datasets for Drug Response Prediction

The success of modern drug response prediction models largely depends on the availability of comprehensive pharmacogenomic datasets containing molecular characteristics of cancer cells and experimentally measured therapeutic responses. Over the past decade, several large-scale initiatives have generated publicly accessible resources that have become fundamental benchmarks for computational drug response prediction research.

These datasets typically integrate molecular information such as gene expression profiles, genomic mutations, copy number variations, methylation patterns, and drug sensitivity measurements. The resulting resources enable researchers to investigate the relationship between cellular characteristics and therapeutic outcomes while facilitating the development of predictive machine learning models.

2.3.1 Genomics of Drug Sensitivity in Cancer (GDSC)

The Genomics of Drug Sensitivity in Cancer (GDSC) database is one of the most widely used pharmacogenomic resources for drug response prediction studies. Developed by the Sanger Institute, GDSC contains molecular profiles of hundreds of cancer cell lines together with experimentally measured responses to numerous anti-cancer compounds.

The dataset includes:

- Gene expression profiles
- Mutation information
- Copy number variation data
- Drug sensitivity measurements
- Molecular drug information

Drug sensitivity within GDSC is commonly represented using IC50 and LN(IC50) values, making the dataset particularly suitable for supervised learning approaches. Due to its large scale and extensive molecular characterization, GDSC has become a standard benchmark for evaluating drug response prediction frameworks.

2.3.2 Cancer Cell Line Encyclopedia (CCLE)

The Cancer Cell Line Encyclopedia (CCLE) is another major pharmacogenomic resource that provides comprehensive molecular characterization of cancer cell lines. CCLE includes genomic, transcriptomic, and mutation data for a wide range of cancer types and has become an essential resource for cancer biology and computational oncology research.

The dataset offers:

- Gene expression profiles
- Mutation annotations
- Copy number variation data
- Cell line metadata
- Multi-omics information

CCLE is frequently integrated with drug response datasets such as GDSC to construct multimodal predictive frameworks capable of learning relationships between cellular features and therapeutic outcomes.

2.3.3 DepMap Project

The Cancer Dependency Map (DepMap) project extends the capabilities of existing pharmacogenomic resources by providing large-scale molecular and functional characterization of cancer cell lines. DepMap integrates information from multiple sources and supports the identification of genetic dependencies and therapeutic vulnerabilities across cancer types.

The project provides:

- Gene expression profiles
- Mutation information
- Functional genomic screening data
- Cell line annotations
- Molecular dependency information

DepMap identifiers are commonly used for dataset integration and cross-referencing across pharmacogenomic resources.

For the present study, data obtained from GDSC, CCLE, and DepMap serve as the primary foundation for constructing the drug response prediction framework. These resources provide the molecular and pharmacological information required to develop task-aware cancer representations, integrate mutation features, and evaluate graph-based drug learning approaches for Non-Small Cell Lung Cancer.

2.4 Traditional Machine Learning Approaches for Drug Response Prediction

Before the widespread adoption of deep learning, drug response prediction primarily relied on traditional machine learning techniques. These methods attempted to establish relationships between molecular characteristics of cancer cells and experimentally measured drug responses using manually engineered features and statistical learning algorithms.

Traditional machine learning models offered several advantages, including interpretability, computational efficiency, and relatively low data requirements. Consequently, they became some of the earliest computational approaches used in pharmacogenomics research. Commonly applied algorithms included Linear Regression, Support Vector Machines (SVMs), Random Forests, Gradient Boosting Machines, and XGBoost.

Linear Regression represents one of the simplest approaches for predicting drug response. The method assumes a linear relationship between molecular features and therapeutic outcomes. Although computationally efficient, Linear Regression struggles to capture the complex nonlinear interactions that characterize biological systems. Cancer progression and drug sensitivity are influenced by intricate interactions among thousands of genes, proteins, and signaling pathways, making purely linear models insufficient for many prediction tasks.

Support Vector Machines were introduced to address some of these limitations by learning nonlinear decision boundaries through kernel functions. SVM-based approaches demonstrated improved predictive performance in several early pharmacogenomic studies. However, their effectiveness decreases as dataset dimensionality increases, particularly when dealing with thousands of genomic features.

Random Forests and Gradient Boosting algorithms further improved predictive performance by combining multiple decision trees into ensemble models. These approaches can model nonlinear relationships and interactions among features more effectively than traditional linear models. Among these methods, XGBoost has emerged as one of the most successful algorithms for structured tabular data and has been widely adopted in biomedical prediction tasks.

Despite their strengths, traditional machine learning approaches face several limitations when applied to drug response prediction. Most importantly, they rely heavily on handcrafted features and feature engineering techniques. High-dimensional biological datasets often contain thousands of genes, many of which exhibit complex nonlinear relationships that are difficult to capture using manually designed representations. Furthermore, traditional machine learning methods struggle to learn hierarchical feature representations directly from raw biological data.

Another limitation involves molecular drug representation. Conventional approaches typically utilize molecular fingerprints or predefined chemical descriptors rather than learning directly from molecular structures. While these representations provide useful information, they may fail to capture important structural relationships within drug molecules.

As pharmacogenomic datasets increased in size and complexity, these limitations motivated researchers to explore deep learning methods capable of automatically learning informative representations from large-scale biological data.

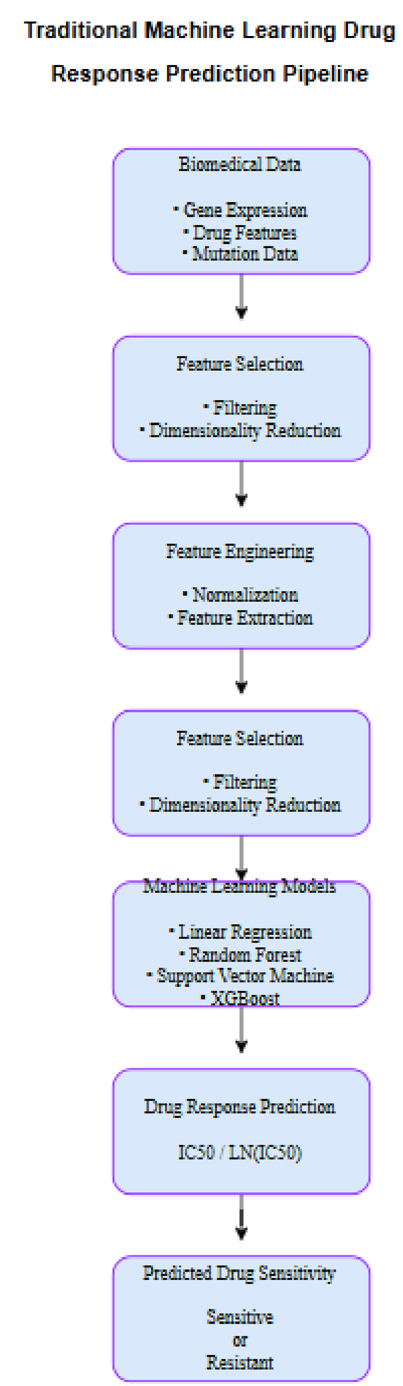

Figure 8:Traditional Machine Learning Drug Response Prediction Pipeline

2.5 Deep Learning for Drug Response Prediction

The emergence of deep learning has significantly transformed computational biology and bioinformatics. Unlike traditional machine learning approaches, deep neural networks can automatically learn hierarchical feature representations from large-scale datasets, enabling them to capture complex nonlinear relationships within biological systems.

Deep learning models consist of multiple interconnected layers that progressively transform raw input data into increasingly abstract representations. This ability to learn features directly from data reduces the dependence on manual feature engineering and allows models to capture intricate biological patterns that may otherwise remain undiscovered.

In the context of drug response prediction, deep learning has demonstrated substantial improvements over traditional machine learning approaches. Researchers have successfully applied Deep Neural Networks (DNNs), Convolutional Neural Networks (CNNs), Autoencoders, Recurrent Neural Networks (RNNs), Attention Mechanisms, and Graph Neural Networks to model relationships between cancer characteristics and therapeutic outcomes.

One major advantage of deep learning is its ability to process heterogeneous biological data. Modern pharmacogenomic datasets often contain multiple data modalities, including gene expression profiles, mutation information, copy number variations, methylation patterns, and drug structures. Deep learning architectures provide flexible mechanisms for integrating these diverse information sources within a unified predictive framework.

2.5.1 Representation Learning Using Autoencoders

Autoencoders are among the most widely used representation learning techniques in computational biology. An autoencoder consists of two primary components:

- Encoder
- Decoder

The encoder compresses high-dimensional input data into a lower-dimensional latent representation, while the decoder attempts to reconstruct the original input from the compressed representation.

The primary objective of an autoencoder is to learn compact latent features that preserve important information while reducing noise and redundancy. This capability is particularly valuable in gene expression analysis, where datasets frequently contain thousands of genes and highly correlated features.

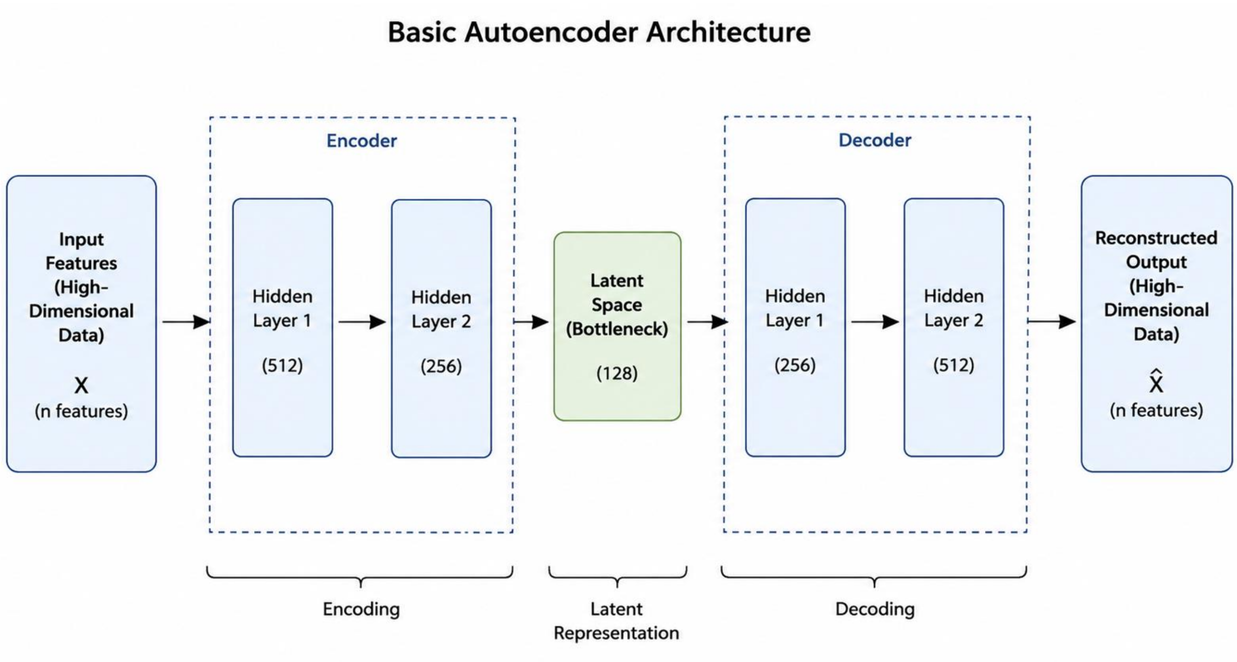

Figure 9:Basic Autoencoder Architecture

Autoencoders have been successfully applied to cancer classification, biomarker discovery, patient stratification, and drug response prediction. By learning compact latent representations, these models reduce dimensionality while preserving biologically meaningful information.

Recent studies have extended traditional autoencoders through task-aware learning strategies. Rather than optimizing only reconstruction quality, task-aware approaches incorporate predictive objectives into the learning process, enabling latent representations to become more relevant for downstream prediction tasks. Such approaches have demonstrated improved performance in several biomedical applications.

2.5.2 Attention Mechanisms in Biomedical Learning

Attention mechanisms were originally introduced within natural language processing but have since been widely adopted across numerous machine learning domains. Attention enables models to selectively focus on the most relevant portions of input data while reducing the influence of less informative features.

In drug response prediction, attention mechanisms have been employed to identify important genes, pathways, and molecular interactions associated with therapeutic response. By assigning different importance weights to input features, attention-based models can improve representation quality and potentially enhance interpretability.

Several recent drug response prediction frameworks have incorporated attention mechanisms to facilitate multimodal data integration. These methods allow models to learn dynamic relationships between cellular characteristics and drug properties rather than relying solely on fixed feature concatenation strategies.

Although attention mechanisms have demonstrated promising results, their effectiveness depends heavily on dataset characteristics, model architecture, and feature representation quality. Consequently, attention-based learning remains an active area of research within computational drug discovery and precision oncology.

2.6 Graph Neural Networks for Drug Representation Learning

Accurate representation of molecular drug structures is one of the most important challenges in drug response prediction. Traditional computational approaches typically represent drugs using molecular fingerprints or predefined chemical descriptors. While these methods capture certain chemical properties, they often fail to preserve the structural relationships present within molecular compounds.

Molecules can naturally be represented as graphs, where atoms correspond to nodes and chemical bonds correspond to edges. This graph-based formulation preserves the structural organization of molecules and enables machine learning algorithms to directly learn from molecular architecture.

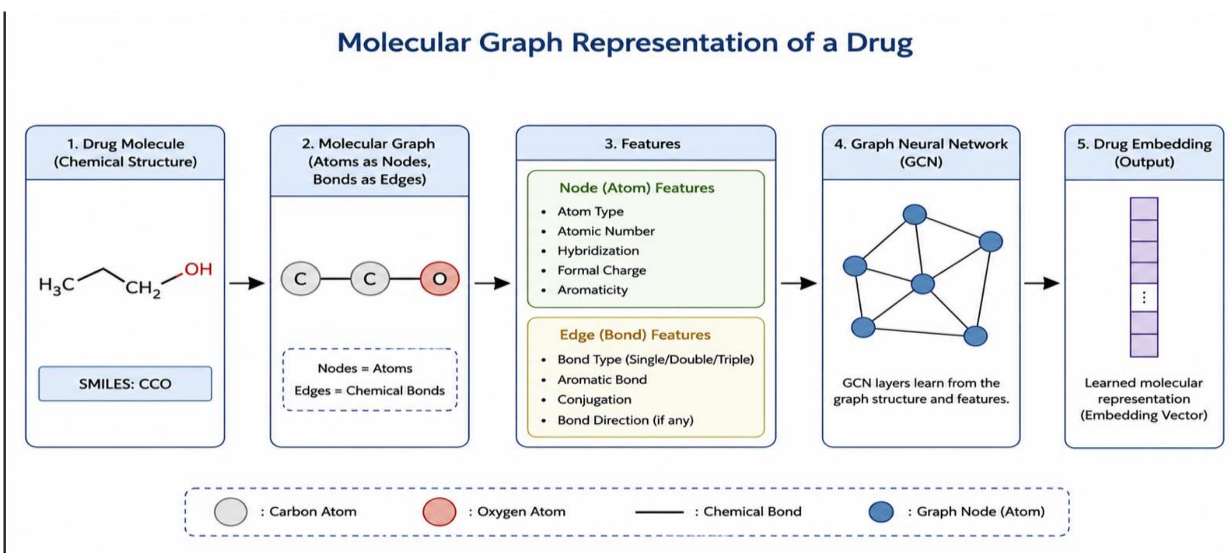

Figure 10:Molecular Graph Representation of a Drug

Graph Neural Networks (GNNs) were developed to perform learning tasks directly on graph-structured data. Unlike conventional neural networks, GNNs propagate information between neighboring nodes, allowing the model to capture both local and global structural properties.
Several GNN variants have been proposed for molecular learning applications.

2.6.1 Graph Convolutional Networks (GCNs)

Graph Convolutional Networks extend traditional convolution operations to graph-structured data. GCNs aggregate information from neighboring nodes and update node representations through multiple graph convolution layers.
In molecular applications, GCNs enable atoms to exchange information with neighboring atoms, thereby learning chemical features that reflect local structural contexts.

2.6.2 Graph Isomorphism Networks (GINs)

Graph Isomorphism Networks were designed to improve the expressive power of graph neural networks. Compared with standard GCNs, GINs are capable of distinguishing a broader range of graph structures and have demonstrated strong performance in molecular property prediction tasks.
Several state-of-the-art drug response prediction frameworks employ GIN-based architectures to generate drug embeddings from molecular graphs.

2.6.3 Graph Transformers

Inspired by the success of Transformer architectures in natural language processing, Graph Transformers incorporate attention mechanisms into graph learning frameworks. These models enable long-range interactions between nodes and improve the ability to capture complex structural dependencies within molecular compounds.
Graph Transformer architectures have recently achieved state-of-the-art performance across various molecular learning benchmarks and are increasingly being adopted in drug discovery research.

Advantages of Graph-Based Drug Learning

Graph Neural Networks offer several advantages over traditional molecular representations:

- Direct learning from molecular structure.
- Preservation of atom-bond relationships.
- Reduced dependence on handcrafted descriptors.
- Improved representation learning capability.
- Enhanced scalability for diverse molecular compounds.

These advantages have made GNNs a cornerstone of modern drug discovery and pharmacogenomic prediction systems. Consequently, many recent drug response prediction frameworks incorporate graph-based drug learning as a fundamental component of their architecture.

Method	**Structural Information**	**Handcrafted Features Required**	**Learning Capability**
Molecular Fingerprints	Limited	Yes	Low
Chemical Descriptors	Limited	Yes	Low
GCN	High	No	High
GIN	High	No	Very High
Graph Transformer	Very High	No	Very High

Table 4:Comparison of Drug Representation Methods

2.7 Review of Existing State-of-the-Art Drug Response Prediction Models

Over the past decade, numerous deep learning frameworks have been proposed to improve the prediction of cancer drug response. These models differ in terms of biological data utilization, drug representation strategies, feature extraction methods, and multimodal integration mechanisms. This section reviews several influential approaches that have significantly contributed to the advancement of computational drug response prediction.

2.7.1 MOLI: Multi-Omics Late Integration

Sharifi-Noghabi et al. proposed MOLI (Multi-Omics Late Integration), a deep learning framework designed to integrate multiple omics data sources for drug response prediction. The model utilizes gene expression, copy number variation, and mutation information as separate inputs and processes each modality through independent neural networks before performing late-stage integration.

The primary objective of MOLI is to exploit complementary biological information from multiple omics sources while preserving modality-specific characteristics. By learning separate feature representations for each data type, MOLI attempts to capture complex biological interactions associated with therapeutic response.

Strengths

- Incorporates multiple omics modalities.
- Learns modality-specific representations.
- Demonstrates improved performance compared to single-modality approaches.
- Provides a flexible framework for multimodal integration.

Limitations

- Does not learn drug representations from molecular structures.
- Relies primarily on cellular information.
- Uses fully connected networks rather than graph-based drug learning.
- Limited ability to capture structural properties of therapeutic compounds.

Although MOLI demonstrated the importance of integrating heterogeneous biological information, the absence of graph-based drug representation learning limits its ability to model complex molecular characteristics of therapeutic agents.

2.7.2 DeepCDR: Hybrid Graph and Genomic Learning

DeepCDR represents one of the earliest attempts to combine graph-based drug representation learning with genomic information for drug response prediction. The framework integrates molecular graph representations of drugs with genomic features derived from cancer cell lines.

The model utilizes Graph Convolutional Networks to learn drug representations directly from molecular structures while simultaneously processing cellular information through deep neural networks. These representations are subsequently fused to predict drug response.

DeepCDR demonstrated that incorporating graph-based drug learning significantly improves predictive performance compared to traditional molecular descriptor approaches.

Strengths

- Introduces graph-based drug representation learning.
- Learns molecular features directly from chemical structures.
- Supports multimodal integration.
- Demonstrates strong predictive performance.

Limitations

- Limited mutation integration.
- Relies heavily on gene expression features.
- Does not employ task-aware representation learning.
- Limited interpretability.

DeepCDR established the importance of graph neural networks within pharmacogenomics and inspired subsequent research on molecular graph learning for drug response prediction.

2.7.3 PaccMann: Attention-Based Drug Sensitivity Prediction

PaccMann (Prediction of Anti-Cancer Compound Sensitivity with Multimodal Attention Networks) introduced attention mechanisms into the drug response prediction domain. The framework combines gene expression information with molecular drug representations while employing attention modules to identify informative biological signals.

The central idea behind PaccMann is that not all genes contribute equally to therapeutic response. By assigning dynamic importance weights to input features, the model attempts to focus on biologically relevant patterns associated with drug sensitivity.

Attention-based learning also improves interpretability by highlighting molecular features that contribute most strongly to predictions.

Strengths

- Incorporates attention mechanisms.
- Improves feature selection through dynamic weighting.
- Supports multimodal learning.
- Provides enhanced interpretability compared to standard deep neural networks.

Limitations

- Does not utilize graph-based molecular learning extensively.
- Limited incorporation of mutation information.
- Attention performance depends heavily on feature quality.
- Computational complexity increases with model size.

PaccMann demonstrated the potential benefits of attention mechanisms for drug response prediction and influenced subsequent research involving attention-based multimodal integration.

2.7.4 GraphDRP: Graph-Based Drug Response Prediction

GraphDRP represents a major advancement in graph-based drug response prediction. Unlike traditional approaches that utilize molecular fingerprints, GraphDRP models drugs directly as molecular graphs and learns drug embeddings through Graph Convolutional Networks.

The framework combines graph-derived drug representations with genomic information obtained from cancer cell lines. By learning structural features directly from molecular architecture, GraphDRP significantly improves the quality of drug representations.

GraphDRP demonstrated that graph-based molecular learning outperforms many conventional descriptor-based approaches and established Graph Neural Networks as a powerful tool for drug response prediction.

Strengths

- Direct molecular graph learning.
- Strong graph representation capability.
- Reduced dependence on handcrafted descriptors.
- Improved predictive performance.

Limitations

- Primarily relies on gene expression features.
- Limited mutation utilization.
- Does not incorporate task-aware cancer embeddings.
- Simplified multimodal fusion strategy.

Although GraphDRP significantly improved drug representation learning, opportunities remained for enhancing cellular feature learning and multimodal integration.

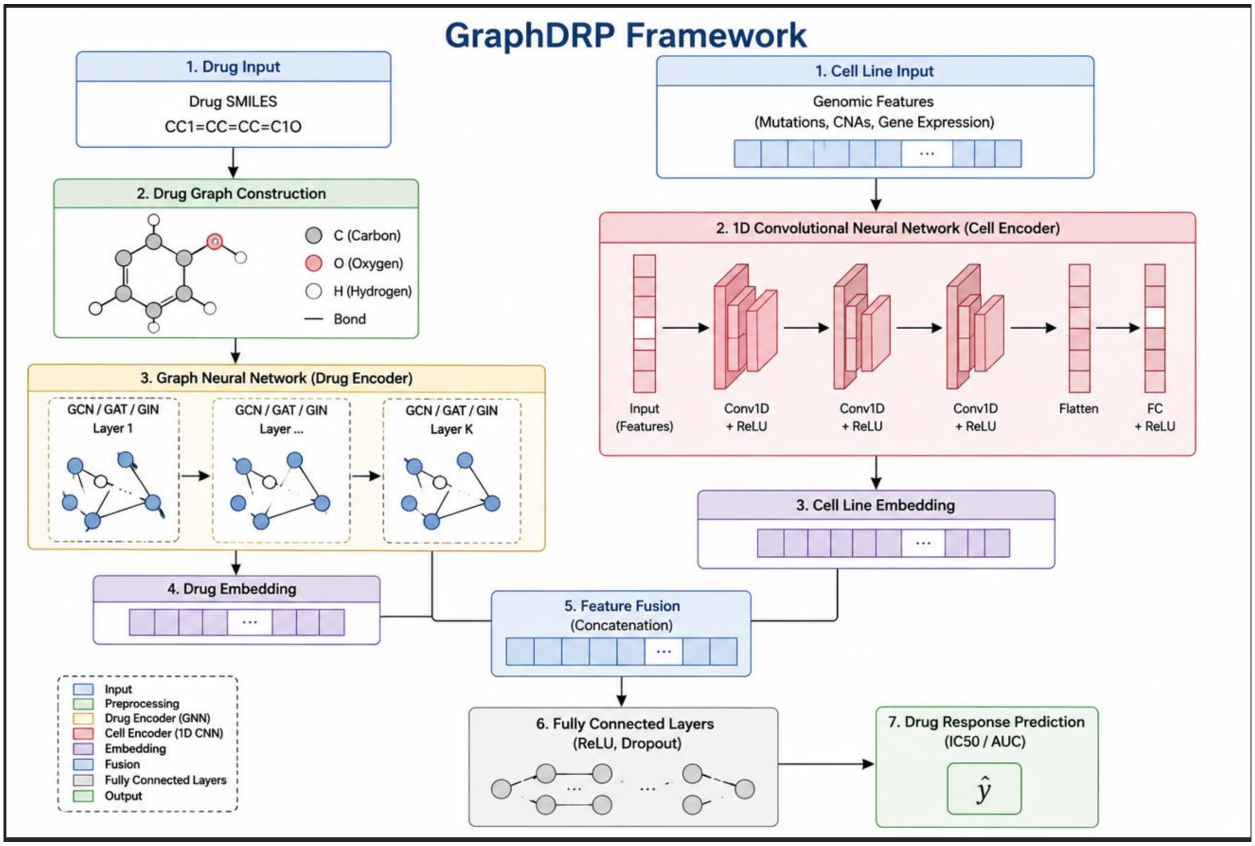

Figure 11:GraphDRP Framework

2.7.5 GPDRP: Graph and Pathway-Based Drug Response Prediction

GPDRP serves as the primary inspiration for the present research. The framework integrates graph-based drug learning with pathway-level cellular information to predict cancer drug response.

In GPDRP, drug molecules are represented as molecular graphs and processed using Graph Isomorphism Networks and Graph Transformer layers. Simultaneously, cellular information is represented through pathway activity scores derived from gene expression data. These representations are combined and passed through fully connected layers to predict LN(IC50) values.

The architecture demonstrated strong predictive performance and highlighted the effectiveness of combining graph-based molecular learning with biologically informed cellular representations.

Strengths

- Advanced graph-based drug representation learning.
- Utilizes Graph Isomorphism Networks.

- Incorporates Graph Transformer layers.
- Strong predictive performance.
- Effective multimodal integration.

Limitations

Despite its strengths, several limitations remain:

- Mutation information is not explicitly integrated.
- Cellular representations are derived from pathway scores rather than learned task-aware embeddings.
- Limited exploration of representation learning techniques for cancer characterization.
- Does not incorporate personalized drug recommendation mechanisms.
- Limited focus on NSCLC-specific modeling.

These limitations motivated the development of the framework proposed in this study

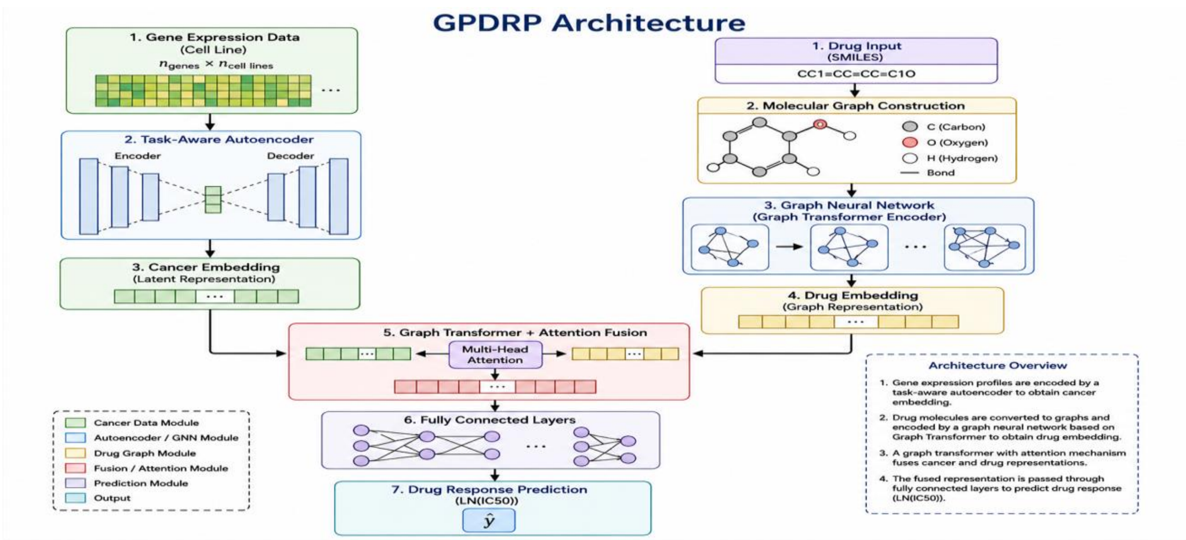

Figure 12:GPDRP Architecture

2.8 Comparative Analysis of Existing Studies

The reviewed studies demonstrate significant progress in drug response prediction through the application of machine learning, deep learning, attention mechanisms, and graph neural networks. However, each framework exhibits distinct strengths and limitations.

Model	**Gene Expression**	**Mutation Data**	**Drug Graph Learning**	**Attention**	**Multi-Omics**	**Task-Aware Learning**
MOLI	✓	✓	✗	✗	✓	✗
DeepCDR	✓	Limited	✓	✗	Partial	✗
PaccMann	✓	✗	Partial	✓	Partial	✗
GraphDRP	✓	✗	✓	✗	✗	✗
GPDRP	✓ (Pathways)	✗	✓	Graph Transformer	✗	✗
Proposed Framework	✓	✓	✓	✓	Partial	✓

Table 5:Comparative Analysis of State-of-the-Art Models

From the comparison, several observations emerge.

First, graph-based drug learning has become increasingly important due to its ability to capture molecular structure directly from chemical graphs. DeepCDR, GraphDRP, and GPDRP all demonstrate the effectiveness of graph neural networks for drug representation learning.

Second, while some studies incorporate mutation information or multi-omics data, few frameworks integrate mutation profiles with learned cancer embeddings and graph-based drug learning simultaneously.

Third, task-aware representation learning remains largely unexplored within existing drug response prediction frameworks. Most approaches utilize predefined biological features or conventional neural network representations without explicitly optimizing cellular embeddings for downstream prediction tasks.

Finally, relatively few studies investigate personalized drug recommendation capabilities beyond simple drug response prediction.

2.9 Research Gap Analysis

Although substantial progress has been achieved in computational drug response prediction, several challenges remain unresolved.

Gap 1: Limited Task-Aware Cancer Representation Learning

Most existing approaches rely on raw gene expression data, pathway activity scores, or manually engineered biological features. Few studies employ task-aware representation learning techniques capable of generating compact cancer embeddings optimized specifically for drug response prediction.

Gap 2: Insufficient Mutation Integration

Genomic mutations play a critical role in determining therapeutic response, particularly in Non-Small Cell Lung Cancer. Despite their biological significance, mutation features remain underutilized in many state-of-the-art frameworks.

Gap 3: Limited NSCLC-Specific Modeling

Many existing studies train predictive models across multiple cancer types. While this approach increases dataset size, it may introduce biological heterogeneity that reduces the model's ability to learn disease-specific response patterns.

Gap 4: Incomplete Multimodal Integration

Current frameworks typically emphasize either cellular representation learning or drug representation learning. Few studies effectively combine task-aware cancer embeddings, mutation information, and graph-based molecular representations within a unified predictive architecture.

Gap 5: Limited Drug Recommendation Functionality

Most drug response prediction systems focus exclusively on regression performance. However, practical precision oncology applications require mechanisms for ranking and recommending candidate drugs based on predicted effectiveness.

Based on these identified gaps, this research proposes a multimodal framework that integrates task-aware cancer embeddings, mutation information, graph-based drug representation learning, and personalized drug recommendation within a unified architecture designed specifically for Non-Small Cell Lung Cancer.

2.10 Chapter Summary

This chapter reviewed the major developments in precision oncology, pharmacogenomics, deep learning, and graph-based drug response prediction. Traditional machine learning approaches, deep learning methods, autoencoders, attention mechanisms, and graph neural networks were discussed. Several influential drug response prediction frameworks, including MOLI, DeepCDR, PaccMann, GraphDRP, and GPDRP, were critically analyzed.

The review identified several limitations within existing studies, including limited mutation integration, insufficient task-aware representation learning, lack of NSCLC-specific modeling, and restricted drug recommendation capabilities. These gaps provide the motivation for the proposed framework presented in the next chapter.

The following chapter describes the research methodology, dataset preparation procedures, task-aware cancer representation learning, mutation integration strategy, graph-based drug learning framework, and the overall architecture developed to address the identified research gaps.

CHAPTER 3

RESEARCH METHODOLOGY

3.1 Introduction

This chapter presents the methodology adopted for the development of the proposed drug response prediction framework. The objective of this research is to predict the sensitivity of Non-Small Cell Lung Cancer (NSCLC) cell lines to anticancer drugs by integrating transcriptomic, genomic, and molecular information within a unified deep learning architecture.

The proposed methodology combines multiple biological data modalities including gene expression profiles, mutation information, and molecular drug structures to model complex drug-cell interactions. The framework employs task-aware representation learning to generate biologically meaningful cancer embeddings, integrates mutation information associated with NSCLC, and utilizes Graph Neural Networks (GNNs) to learn molecular representations directly from drug structures.

The methodology consists of several stages including data collection, preprocessing, cancer representation learning, mutation integration, molecular graph construction, graph neural network training, multimodal fusion, drug response prediction, and personalized drug recommendation. Each stage was carefully designed to improve predictive performance while preserving biological relevance.

The overall workflow begins with the acquisition of pharmacogenomic datasets from publicly available resources including the Genomics of Drug Sensitivity in Cancer (GDSC), Cancer Cell Line Encyclopedia (CCLE), and Cancer Dependency Map (DepMap). Following preprocessing and NSCLC-specific filtering, gene expression and mutation information are processed to construct cellular representations. Drug molecules are converted into graph structures using molecular SMILES representations and subsequently processed through Graph Neural Networks. The learned cellular and molecular embeddings are fused within a multimodal prediction framework to estimate drug response in terms of LN(IC50) values.

Finally, the trained model is utilized to evaluate candidate drugs for individual cancer cell lines and generate personalized drug recommendations based on predicted therapeutic effectiveness.

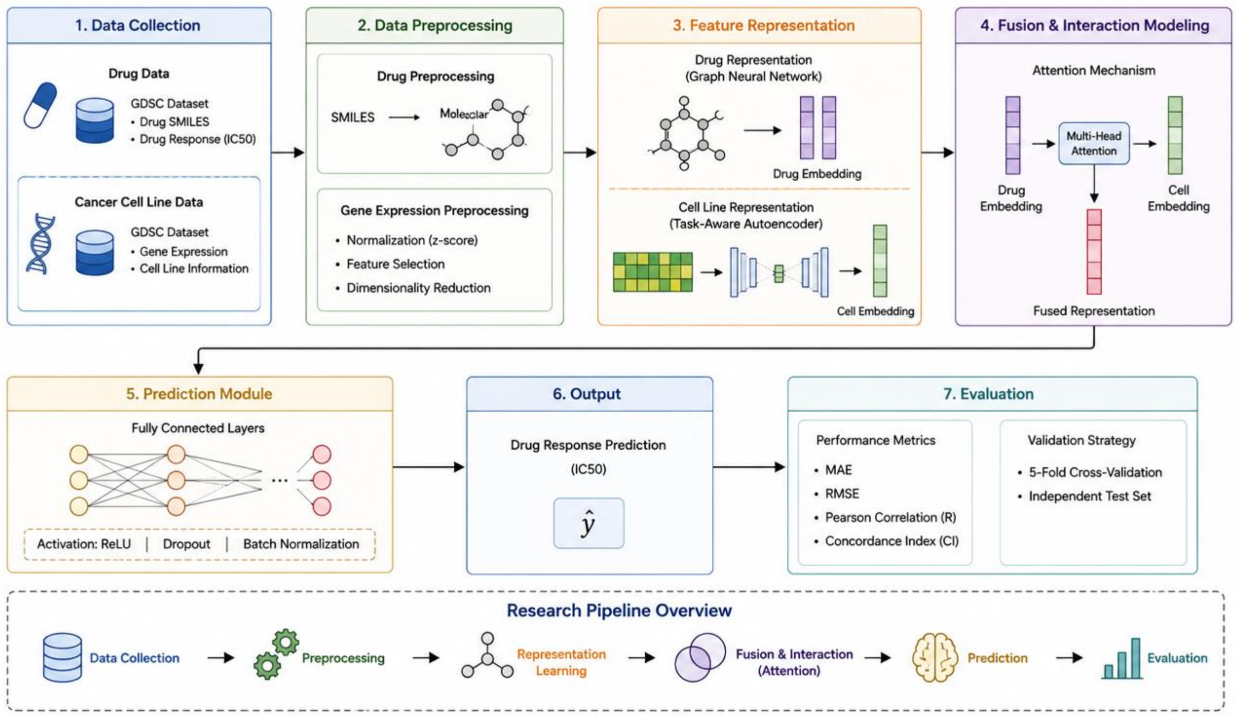

Figure 13:Overall Research Framework

3.2 Research Framework

The proposed framework was designed to predict drug response by integrating cellular and molecular information within a unified deep learning architecture. Unlike conventional machine learning methods that depend heavily on handcrafted features, the proposed methodology employs representation learning techniques capable of automatically extracting informative biological and chemical patterns from raw data.

The framework consists of five major components:

1. Data Collection and Integration
2. Cancer Representation Learning
3. Mutation Integration
4. Graph-Based Drug Representation Learning
5. Drug Response Prediction and Recommendation

Initially, pharmacogenomic datasets were collected from multiple publicly available repositories. Gene expression profiles, mutation information, and drug response measurements were integrated using common identifiers to establish relationships between cell lines and therapeutic compounds. Following preprocessing, a task-aware learning framework was employed to generate compact cancer embeddings from high-dimensional gene expression profiles. Mutation information was subsequently integrated with these learned representations to provide additional biological context regarding genomic alterations associated with NSCLC.

Simultaneously, molecular drug structures represented using SMILES strings were transformed into graph structures where atoms were represented as nodes and chemical bonds were represented as edges. Graph Neural Networks were then used to learn molecular embeddings directly from these graph representations.

The resulting cellular and molecular embeddings were fused and processed through a regression network to predict LN(IC50) values. Predicted responses were subsequently utilized to rank candidate drugs and generate personalized treatment recommendations.

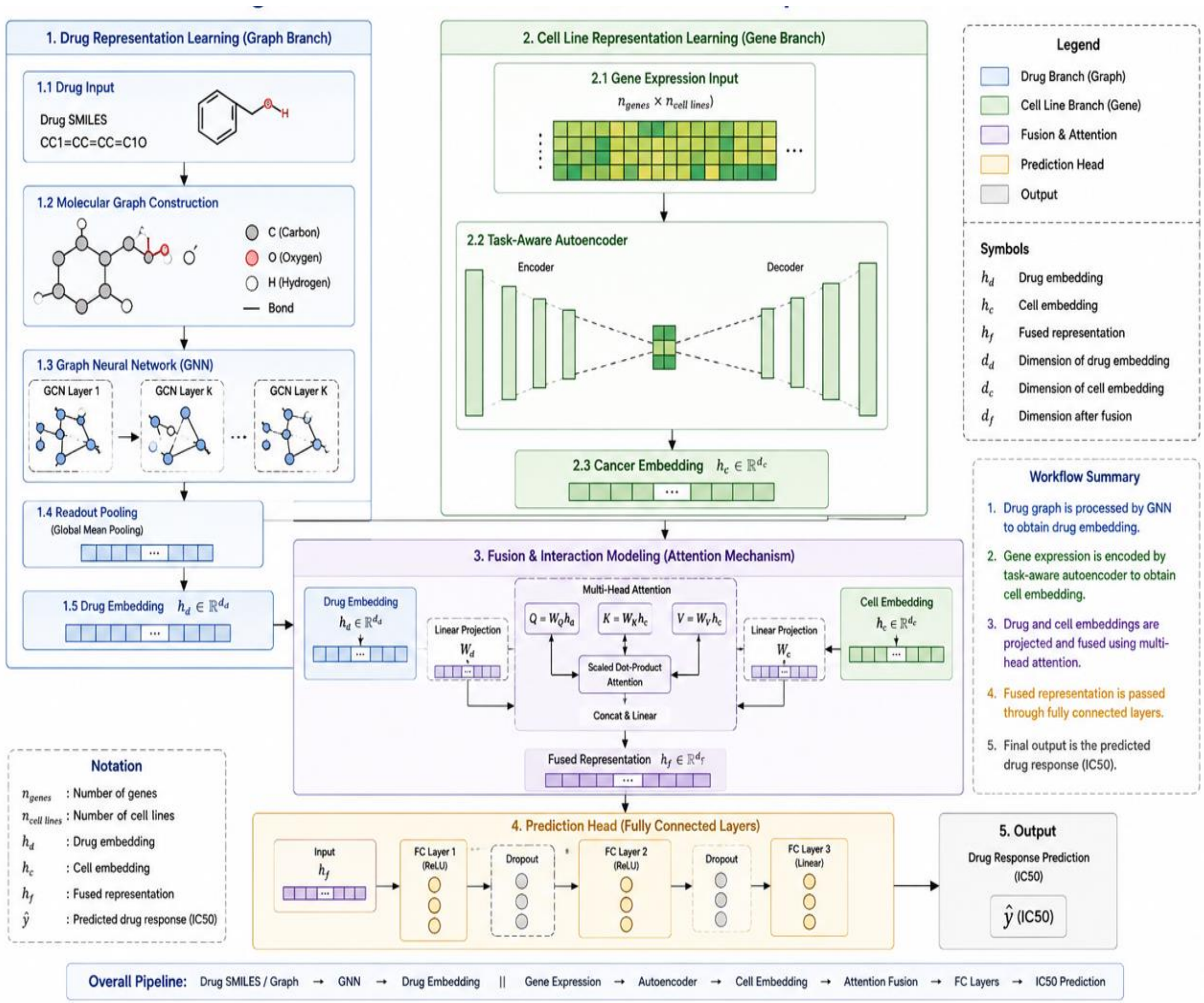

Figure 14:Detailed Architecture of the Proposed Framework

3.3 Data Collection and Sources

The proposed framework utilizes multiple publicly available pharmacogenomic resources that provide complementary biological and pharmacological information. These datasets collectively supply the transcriptomic, genomic, molecular, and therapeutic response information required for model development.

3.3.1 Genomics of Drug Sensitivity in Cancer (GDSC)

The Genomics of Drug Sensitivity in Cancer (GDSC) database was used as the primary source of drug response information. GDSC contains experimentally measured responses of cancer cell lines to a large collection of anticancer compounds and serves as one of the most widely used benchmarks for drug response prediction research.

Drug sensitivity within GDSC is represented using LN(IC50), which indicates the concentration of a drug required to inhibit cellular growth by 50 percent. Lower LN(IC50) values correspond to greater drug sensitivity, whereas higher values indicate reduced therapeutic effectiveness.

The final dataset used in this research contained:

- 229 anticancer drugs
- 19,207 drug-response observations

The GDSC dataset also provided molecular structure information through canonical SMILES representations that were later converted into graph structures for Graph Neural Network learning.

3.3.2 Cancer Cell Line Encyclopedia (CCLE)

Mutation information was obtained from the Cancer Cell Line Encyclopedia (CCLE). CCLE provides comprehensive genomic characterization of cancer cell lines and contains mutation annotations across numerous cancer-associated genes.

For this study, mutation information was extracted for sixteen clinically relevant NSCLC-associated genes. These mutations were encoded and integrated into the cellular representation learning pipeline to enhance biological characterization.

The selected genes include:

- TP53
- EGFR
- KRAS
- STK11
- KEAP1

- BRAF
- PIK3CA
- ALK
- ROS1
- MET
- ERBB2
- NF1
- RB1
- SMARCA4
- CDKN2A
- PTEN

These genes are frequently implicated in lung cancer progression, targeted therapy response, and drug resistance mechanisms.

3.3.3 Cancer Dependency Map (DepMap)

Gene expression profiles were obtained from the Cancer Dependency Map (DepMap) project. DepMap provides transcriptomic measurements for a large collection of cancer cell lines and serves as an important resource for computational oncology research.

After preprocessing and filtering, the final gene expression dataset contained:

- 92 NSCLC cell lines
- 8,147 gene expression features

The processed dataset was stored as:

FINAL_NSCLC_EXPRESSION.csv

and served as the primary source of transcriptomic information for cancer representation learning.

Dataset	**Purpose**	**Information Provided**
GDSC	Drug Response	LN(IC50), Drug Structures
CCLE	Mutation Information	Genomic Alterations
DepMap	Gene Expression	Transcriptomic Profiles

Table 6:Summary of Datasets Used in the Study

3.4 Data Preprocessing

Data preprocessing is an essential stage in pharmacogenomic modeling because raw biological datasets frequently contain heterogeneous information, missing values, inconsistent identifiers, and irrelevant features. Several preprocessing procedures were therefore performed to ensure compatibility among datasets and improve data quality.

3.4.1 Selection of NSCLC Cell Lines

The original pharmacogenomic resources contain cell lines originating from multiple cancer types. Since the objective of this study is to develop an NSCLC-specific drug response prediction framework, only Non-Small Cell Lung Cancer cell lines were retained.

Cell line metadata was examined to identify lung cancer samples. Small Cell Lung Cancer (SCLC) cell lines were subsequently removed to reduce biological heterogeneity and improve disease-specific modeling.

Following filtering, the final dataset contained:

92 NSCLC Cell Lines

Restricting the study to NSCLC allows the model to learn disease-specific molecular characteristics and therapeutic response patterns rather than attempting to generalize across multiple cancer types.

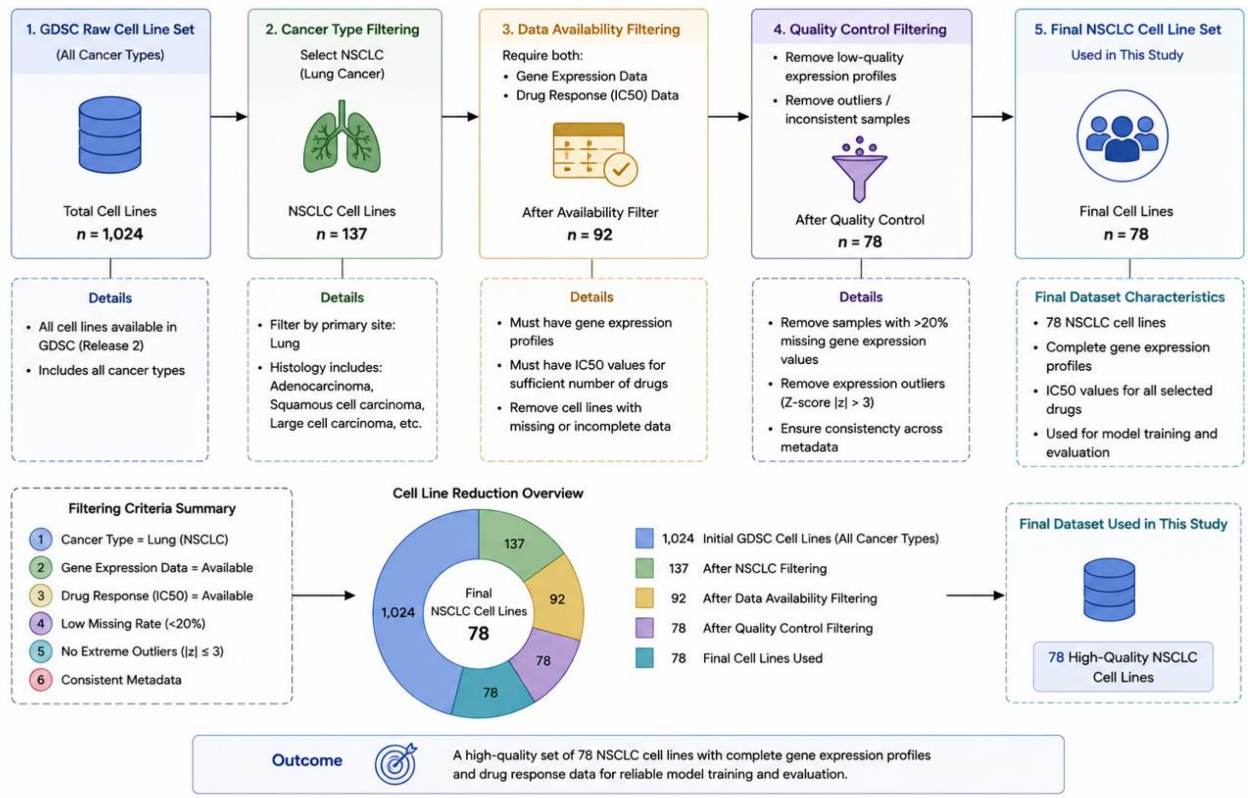

Figure 15:NSCLC Cell Line Filtering Process

3.4.2 Gene Expression Processing

Gene expression data obtained from DepMap underwent preprocessing prior to model development.

The preprocessing procedure included:

- Verification of cell line identifiers
- Removal of irrelevant metadata
- Missing value inspection
- Data consistency validation
- Formatting and normalization

Following preprocessing, the final gene expression matrix contained:

92 Cell Lines × 8,147 Gene Features

The processed dataset was stored as:

FINAL_NSCLC_EXPRESSION.csv

and subsequently used for cancer representation learning.

3.4.3 Mutation Processing

Mutation information obtained from CCLE was transformed into a binary mutation matrix representing the presence or absence of mutations within selected NSCLC-associated genes.

Each mutation feature was encoded as:

1 = Mutation Present

0 = Mutation Absent

The resulting mutation matrix contained sixteen genomic features corresponding to clinically significant NSCLC biomarkers.

These mutation features were later integrated with task-aware cancer embeddings to generate enriched cellular representations.

3.4.4 Dataset Integration

The final preprocessing stage involved integrating gene expression data, mutation information, drug response measurements, and molecular drug information into a unified dataset.

Datasets were linked using common identifiers including:

- ModelID
- Drug Name

Following integration, the final dataset contained:

- 92 NSCLC Cell Lines
- 229 Drugs
- 19,207 Drug-Response Samples

This integrated dataset served as the foundation for cancer representation learning, graph-based drug modeling, and drug response prediction.

3.5 Cancer Representation Learning

Gene expression datasets used in pharmacogenomics are inherently high-dimensional and often contain substantial redundancy, noise, and correlated features. Directly utilizing thousands of gene expression measurements as input to predictive models may increase computational complexity, introduce overfitting, and reduce model generalization. Therefore, an effective representation learning strategy is required to generate compact and informative cellular representations.

To address this challenge, an Autoencoder-based representation learning framework was employed. Autoencoders are neural networks specifically designed to learn compressed latent representations of high-dimensional data while preserving essential information. The architecture consists of two primary components: an encoder and a decoder. The encoder transforms the original gene expression profile into a lower-dimensional latent representation, whereas the decoder reconstructs the original input from the compressed representation.

The encoder learns a compact representation that captures the most informative molecular characteristics of cancer cells. This process reduces dimensionality while retaining biologically meaningful patterns present within the transcriptomic data.

The input to the representation learning framework consisted of 8,147 gene expression features obtained from the processed NSCLC dataset. During training, the Autoencoder minimized reconstruction error between the original gene expression profile and its reconstructed output. This encouraged the model to learn latent representations capable of preserving important biological information while eliminating redundant features.

The resulting cancer embeddings provided a compact representation of each NSCLC cell line and served as the foundation for subsequent task-aware learning and drug response prediction experiments.

Reconstructed Gene Expression

The learned embeddings significantly reduced the dimensionality of the original dataset while preserving biologically relevant information required for downstream analysis.

3.6 Task-Aware Representation Learning

Although conventional Autoencoders effectively reduce dimensionality, they optimize only reconstruction quality and are not explicitly trained to improve predictive performance. As a result, the generated latent features may contain information useful for reconstruction but less relevant for drug response prediction.

To overcome this limitation, a Task-Aware Representation Learning strategy was implemented. The objective of task-aware learning is to guide the representation learning process toward features that are directly useful for predicting therapeutic response.

Instead of optimizing only reconstruction loss, the encoder was coupled with a prediction objective associated with drug response estimation. This additional objective encourages the latent representations to capture information that contributes to both accurate reconstruction and effective drug response prediction.

The task-aware learning process enables the model to learn biologically meaningful embeddings that are optimized for downstream predictive tasks. Consequently, the generated representations become more discriminative and informative than those produced by a conventional Autoencoder. The output of this stage consisted of a 256-dimensional embedding for each NSCLC cell line. These embeddings summarize complex transcriptomic patterns within a compact feature vector and serve as the primary cellular representation within the proposed framework.

The generated embeddings were stored in:

TASK_AWARE_CANCER_EMBEDDINGS.csv

and subsequently utilized during mutation integration and multimodal drug response prediction.

By incorporating predictive objectives into the learning process, the resulting embeddings become more suitable for drug response prediction compared to traditional unsupervised representations.

3.7 Mutation Integration

Genomic mutations are among the most important determinants of therapeutic response in Non-Small Cell Lung Cancer. Mutations occurring within genes such as EGFR, KRAS, TP53, ALK, and STK11 influence cellular signaling pathways, targeted therapy sensitivity, drug resistance mechanisms, and disease progression. Therefore, incorporating mutation information can provide valuable biological context beyond transcriptomic measurements alone.

To enhance cellular representation, mutation information was integrated with the task-aware cancer embeddings generated during the previous stage. Sixteen clinically relevant NSCLC-associated genes were selected based on their established significance within lung cancer research and targeted treatment strategies.

The selected genes include:

- TP53
- EGFR
- KRAS
- STK11
- KEAP1
- BRAF
- PIK3CA
- ALK

- ROS1
- MET
- ERBB2
- NF1
- RB1
- SMARCA4
- CDKN2A
- PTEN

Mutation status for each gene was represented using binary encoding, where:

1 = Mutation Present

0 = Mutation Absent

Each NSCLC cell line therefore possessed a mutation vector consisting of sixteen genomic features.

To construct the final cellular representation, the mutation vector was concatenated with the task-aware embedding generated from gene expression data.

The integration process can be represented as:

Final Cell Features = Task-Aware Embedding + Mutation Features

Where:

- Task-Aware Embedding Dimension = 256
- Mutation Features = 16
- Final Cell Representation = 272 Features

The resulting cellular representation combines transcriptomic and genomic information within a unified feature vector capable of capturing both gene expression patterns and mutation-driven biological characteristics.

The final integrated cellular representations were stored in:

TASK_AWARE_CELL_FEATURES.csv

Dataset Shape:

92 × 273

where the dataset contains:

- 92 NSCLC Cell Lines
- 272 Cellular Features

- 1 ModelID Column

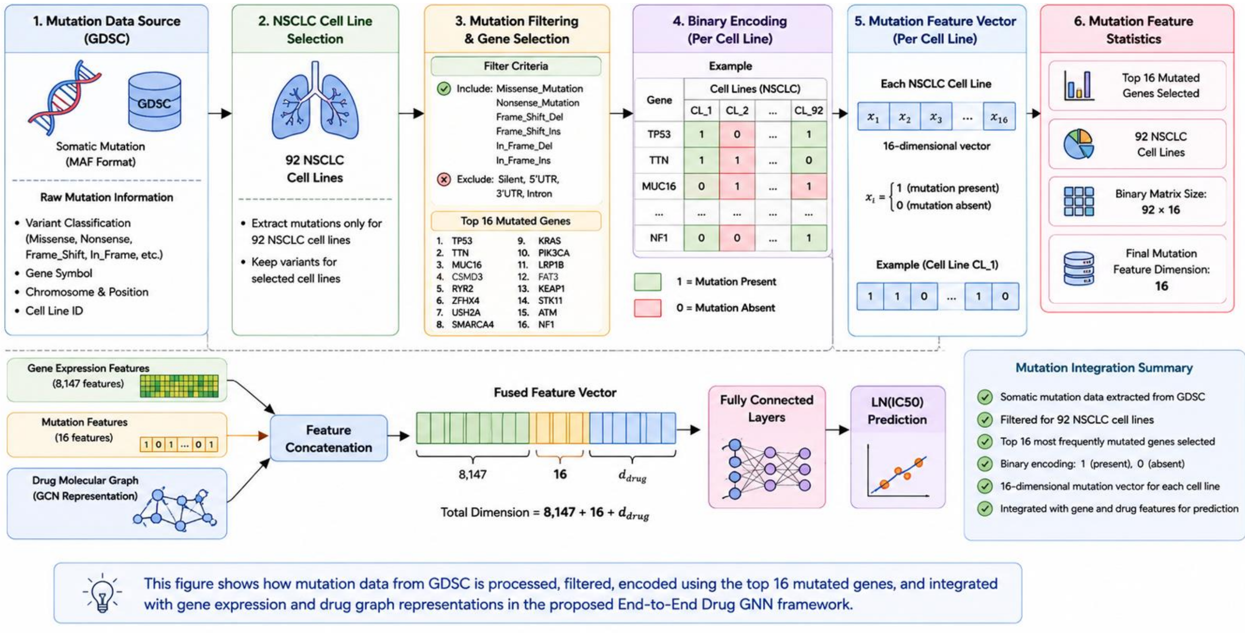

Figure 16:Mutation Integration Process

The integrated feature set serves as the cellular input to the final End-to-End Drug Graph Neural Network framework.

3.8 Drug Graph Construction

Drug molecules contain complex structural information that plays a fundamental role in determining therapeutic effectiveness. Traditional computational approaches often represent drugs using molecular fingerprints or handcrafted chemical descriptors. Although these representations provide useful information, they may fail to capture important structural relationships present within molecular compounds.

To preserve molecular structure, drugs were represented as graphs. Graph-based representations provide a natural way of modeling molecules because atoms and chemical bonds can be directly mapped to graph components.

In the proposed framework:

- Atoms are represented as nodes.
- Chemical bonds are represented as edges.

Drug molecular structures were obtained from the GDSC dataset in the form of canonical SMILES strings. These SMILES representations were converted into molecular graphs using the RDKit cheminformatics library.

The processed drug dataset contained:

229 Drugs

stored in:

GDSC_FULL_SMILES.csv

For each molecular graph, node-level features were extracted to characterize the chemical properties of individual atoms.

The extracted atomic features included:

- Atomic Number
- Atom Degree
- Formal Charge
- Hybridization State
- Aromaticity Indicator

These features collectively describe the chemical environment surrounding each atom and provide informative inputs for graph neural network learning.

By transforming drug molecules into graph structures, the model can directly learn molecular representations from chemical architecture rather than relying on manually engineered descriptors.

3.9 End-to-End Drug Graph Neural Network

Following the generation of cellular representations and molecular graph structures, an End-to-End Drug Graph Neural Network (Drug GNN) was developed to predict drug response. The objective of the model is to learn meaningful representations from both cancer cell features and molecular drug structures while capturing complex interactions between them.

Unlike traditional machine learning approaches that rely on handcrafted descriptors, the proposed architecture learns feature representations directly from biological and molecular data. This enables the model to discover complex nonlinear relationships associated with drug sensitivity and resistance.

The proposed framework consists of three major components:

1. Cell Representation Branch
2. Drug Graph Learning Branch

3. Prediction Network

The overall architecture is illustrated in Figure 3.8.

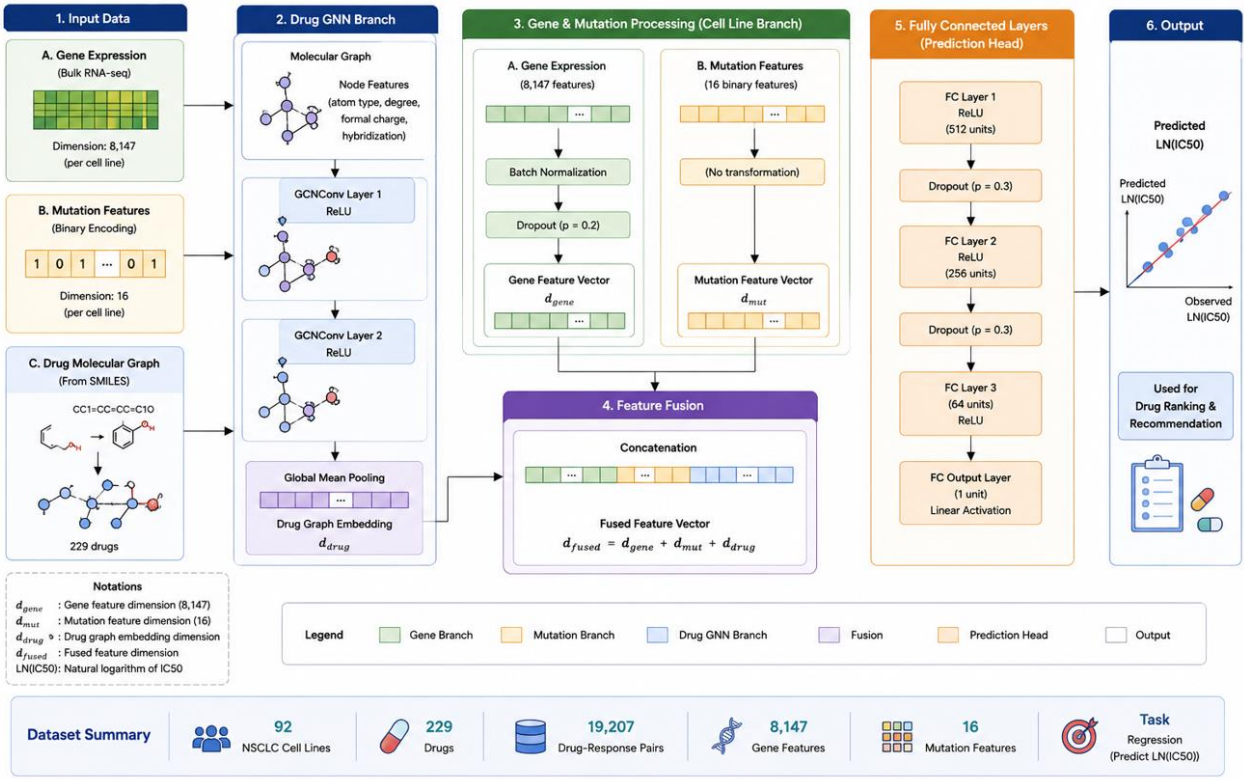

Figure 17:Proposed End-to-End Drug GNN Architecture

3.9.1 Cell Representation Branch

The cellular branch processes the integrated feature representation generated during the previous stages. Each cell line is represented using a 272-dimensional feature vector consisting of:

- 256 Task-Aware Features
- 16 Mutation Features

The cellular representation is passed through a fully connected neural network to generate a dense cellular embedding.

Input Dimension:

272

Transformation:

272 → 256

Activation Function:

ReLU

Output:

256-Dimensional Cell Embedding

The purpose of this branch is to transform heterogeneous biological information into a compact feature representation suitable for multimodal fusion.

3.9.2 Drug Graph Learning Branch

The drug branch is responsible for learning molecular representations directly from graph-structured drug molecules.

Each drug is represented as a molecular graph where:

- Nodes represent atoms.
- Edges represent chemical bonds.

Graph learning is performed using Graph Convolutional Networks (GCNs). Multiple graph convolution layers enable information exchange among neighboring atoms, allowing the network to learn local and global structural properties of the molecule.

The architecture consists of:

First Graph Convolution Layer:

$5 \rightarrow 64$

ReLU Activation

Second Graph Convolution Layer:

$64 \rightarrow 128$

ReLU Activation

Following graph convolutions, Global Mean Pooling is applied to aggregate node-level information into a graph-level representation.

The pooled representation is subsequently passed through a fully connected layer:

$128 \rightarrow 256$

Output:

256-Dimensional Drug Embedding

This embedding captures structural characteristics of the drug molecule and serves as the molecular representation used during fusion.

3.9.3 Multimodal Fusion

Following representation learning, cellular and molecular embeddings are combined to create a unified feature representation.

The fusion process utilizes feature concatenation:

Cell Embedding:

256 Features

Drug Embedding:

256 Features

Concatenated Representation:

512 Features

This fused representation contains complementary biological and chemical information and serves as the input to the prediction network.

The fusion operation can be expressed as:

Fusion Vector = [Cell Embedding || Drug Embedding]

Resulting Dimension:

512

The fused feature vector enables the model to jointly analyze cancer-specific and drug-specific characteristics when predicting therapeutic response.

3.9.4 Drug Response Prediction Head

The prediction network receives the fused representation and estimates the corresponding LN(IC50) value.

The prediction head consists of multiple fully connected layers designed to learn nonlinear interactions between cellular and molecular features.

Layer Configuration:

512 → 512

ReLU Activation

Dropout (0.3)

↓

512 → 256

ReLU Activation

Dropout (0.3)

↓

256 → 128

ReLU Activation

↓

128 → 1

Output:

Predicted LN(IC50)

The final output represents the predicted drug sensitivity of a specific NSCLC cell line to a specific therapeutic compound.

Lower predicted LN(IC50) values indicate greater drug effectiveness, whereas higher values suggest reduced sensitivity.

3.10 Personalized Drug Recommendation Framework

Although drug response prediction provides valuable information regarding therapeutic effectiveness, practical precision oncology applications require mechanisms for recommending candidate treatments.

To address this requirement, a personalized drug recommendation framework was developed on top of the trained Drug GNN model.

The recommendation process begins by selecting a target NSCLC cell line. The cellular representation of the selected cell line is combined with every available drug within the dataset. The trained model then predicts LN(IC50) values for all candidate drugs.

The prediction workflow can be summarized as follows:

Selected Cell Line

↓

Evaluate Drug 1

↓

Predict LN(IC50)

↓

Evaluate Drug 2

↓

Predict LN(IC50)

↓

...

↓

Evaluate Drug 229

↓

Predict LN(IC50)

Following prediction, drugs are sorted according to their predicted LN(IC50) values.

Lower predicted values correspond to greater sensitivity and therefore higher recommendation priority.

The final output consists of a ranked list of candidate drugs, enabling the identification of potentially effective therapeutic compounds for a given cancer profile.

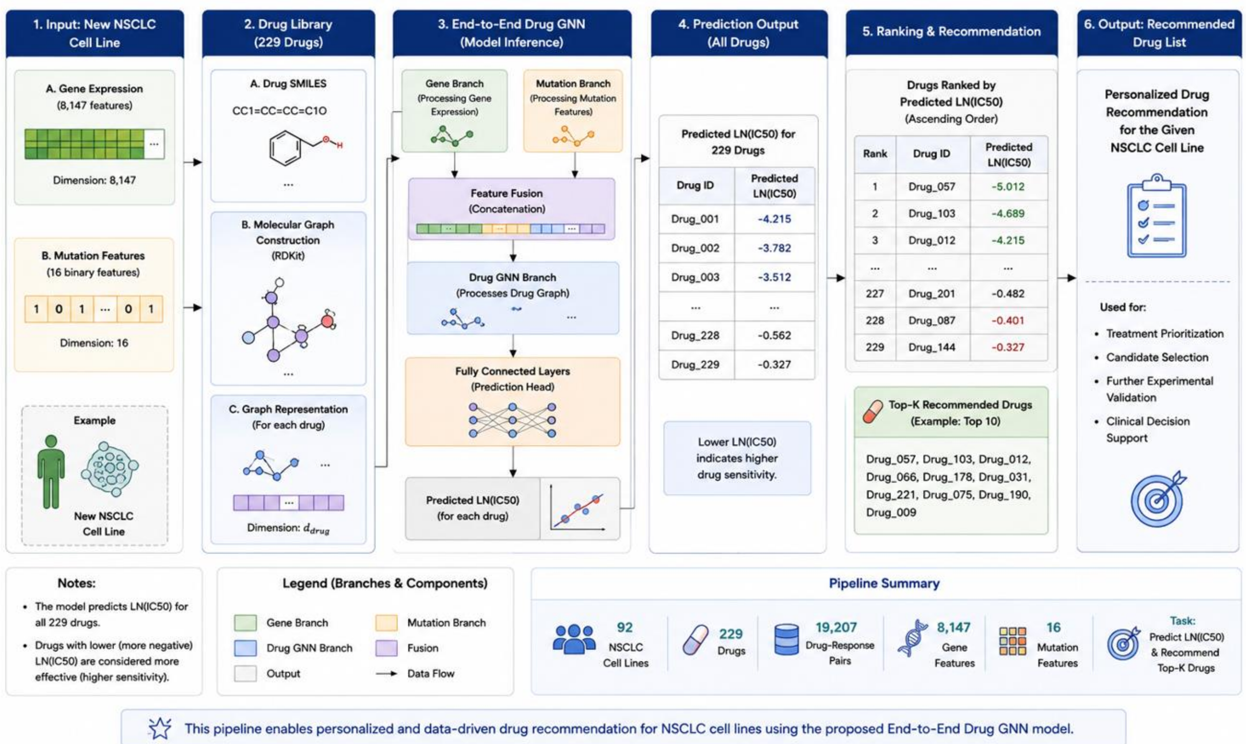

Figure 18:Drug Recommendation Pipeline

Example recommendations generated by the proposed framework include compounds such as:

- Docetaxel
- Romidepsin

- Paclitaxel
- Vinorelbine
- Vinblastine

These recommendations demonstrate the practical applicability of the proposed framework beyond simple regression prediction.

3.11 Experimental Setup

All experiments were conducted using the Google Colaboratory platform with GPU acceleration enabled. The implementation was developed using Python and several open-source machine learning libraries.

The software environment included:

- Python
- PyTorch
- PyTorch Geometric
- RDKit
- NumPy
- Pandas
- Scikit-learn

Graph neural network operations were implemented using PyTorch Geometric, while RDKit was utilized for molecular graph generation and feature extraction.

Parameter	**Value**
Framework	PyTorch
Graph Library	PyTorch Geometric
Molecular Processing	RDKit
Optimizer	Adam
Learning Rate	0.0001
Batch Size	64
Loss Function	MSE
Epochs	230

Table 7:Experimental Configuration

Device GPU (CUDA)

The dataset was divided using an 80:20 train-test split.

Training Samples:

15,365

Testing Samples:

3,842

Total Samples:

19,207

The model was trained for a total of 230 epochs to ensure convergence and stable learning.

3.12 Evaluation Metrics

The performance of the proposed framework was evaluated using four widely adopted regression metrics.

3.12.1 Root Mean Square Error (RMSE)

RMSE measures the average magnitude of prediction errors and penalizes larger errors more heavily.

Formula:

$RMSE = \sqrt{[(1/n) \Sigma(y_i - \hat{y}_i)^2]}$

where:

y_i = Actual Value

$\hat{y}_i$ = Predicted Value

n = Number of Samples

Lower RMSE values indicate better predictive performance.

3.12.2 Mean Absolute Error (MAE)

MAE measures the average absolute difference between actual and predicted values.

Formula:

$MAE = (1/n) \Sigma |y_i - \hat{y}_i|$

Lower MAE values indicate improved prediction accuracy.

3.12.3 Coefficient of Determination (R^2)

R^2 evaluates the proportion of variance explained by the model.

Formula:

$R^2 = 1 - (SS\_res / SS\_tot)$

Higher R^2 values indicate stronger predictive capability.

3.12.4 Pearson Correlation Coefficient (PCC)

Pearson Correlation measures the strength of linear association between actual and predicted responses.

Formula:

$PCC = Cov(X,Y) / (\sigma X \times \sigma Y)$

Values range from:

-1 to +1

Higher positive values indicate stronger agreement between predictions and actual responses.

Pearson Correlation is widely considered one of the most important evaluation metrics within drug response prediction research.

3.13 Chapter Summary

This chapter presented the methodology adopted for the development of the proposed drug response prediction framework. Data were collected from GDSC, CCLE, and DepMap and subsequently preprocessed to construct an NSCLC-specific pharmacogenomic dataset. Task-aware representation learning was employed to generate compact cancer embeddings, while mutation information was integrated to enhance biological characterization.

Drug molecules were represented as molecular graphs and processed using Graph Neural Networks to learn molecular embeddings directly from chemical structures. Cellular and molecular representations were fused within an End-to-End Drug GNN architecture to predict LN(IC50) values. Finally, a personalized drug recommendation framework was developed to rank candidate therapies according to predicted effectiveness.

The following chapter describes the implementation details, software tools, model development process, and experimental procedures used to construct the proposed framework.

Chapter 04

4.1 Introduction

This chapter presents the implementation details of the proposed drug response prediction framework. The chapter describes the software tools, development environment, libraries, model implementation procedures, and experimental configurations used throughout the study. Particular emphasis is placed on the implementation of cancer representation learning, mutation integration, graph-based drug representation learning, and the End-to-End Drug Graph Neural Network developed for drug response prediction.

The implementation was carried out using modern deep learning and graph learning frameworks capable of handling large-scale pharmacogenomic datasets. Multiple experiments were conducted to evaluate different model architectures and identify the most effective framework for predicting drug response in Non-Small Cell Lung Cancer.

The chapter also describes the computational environment, hyperparameter configurations, training procedures, and evaluation protocols employed during experimentation.

4.2 Development Environment

The proposed framework was implemented using Python due to its extensive support for machine learning, deep learning, bioinformatics, and cheminformatics applications.

Model development and experimentation were conducted using Google Colaboratory, which provided access to cloud-based GPU resources for efficient model training and evaluation.

The implementation environment consisted of several open-source libraries that collectively supported data preprocessing, representation learning, graph neural network development, molecular graph construction, and model evaluation.

Tool / Library	**Purpose**
Python	Programming Language
Google Colab	Development Environment
Pandas	Data Manipulation
NumPy	Numerical Computation
Scikit-Learn	Evaluation Metrics

TensorFlow/Keras	Autoencoder Development
PyTorch	Deep Learning
PyTorch Geometric	Graph Neural Networks
RDKit	Molecular Processing
Matplotlib	Visualization
Seaborn	Data Visualization

Table 8:Software and Libraries Used

The combination of these tools enabled efficient implementation of both conventional neural networks and graph-based learning architectures.

4.3 Dataset Implementation

Following data collection and preprocessing, the datasets were organized into a structured format suitable for model development.

The final implementation utilized three primary data sources:

Gene Expression Dataset

File:

FINAL_NSCLC_EXPRESSION.csv

Contents:

- 92 NSCLC Cell Lines
- 8,147 Gene Expression Features

Purpose:

Cancer representation learning.

Mutation Dataset

File:

CCLE_mutations.csv

Processed Output:

MUTATION_MATRIX.csv

Contents:

- 16 Mutation Features

Purpose:

Genomic characterization of NSCLC cell lines.

Drug Dataset

File:

GDSC_FULL_SMILES.csv

Contents:

- 229 Drugs
- Canonical SMILES Strings

Purpose:

Graph-based drug representation learning.

Drug Response Dataset

File:

FINAL_NSCLC_GDSC.csv

Contents:

- 19,207 Drug Response Samples
- LN(IC50) Values

Purpose:

Supervised learning target variable.

4.4 Autoencoder Implementation

The first stage of implementation involved constructing a deep Autoencoder for cancer representation learning.

The Autoencoder was implemented using TensorFlow and Keras. The architecture consisted of multiple fully connected layers arranged symmetrically around a bottleneck layer.

The encoder progressively reduced the dimensionality of the gene expression data from 8,147 features to a compact 256-dimensional latent representation.

The implementation included:

- Dense Layers
- Batch Normalization

- Dropout Regularization
- ReLU Activation Functions

Batch Normalization was incorporated to stabilize learning and accelerate convergence. Dropout layers with a rate of 0.3 were used to reduce overfitting and improve model generalization.
The Autoencoder was trained using the Adam optimizer and Mean Squared Error loss function.

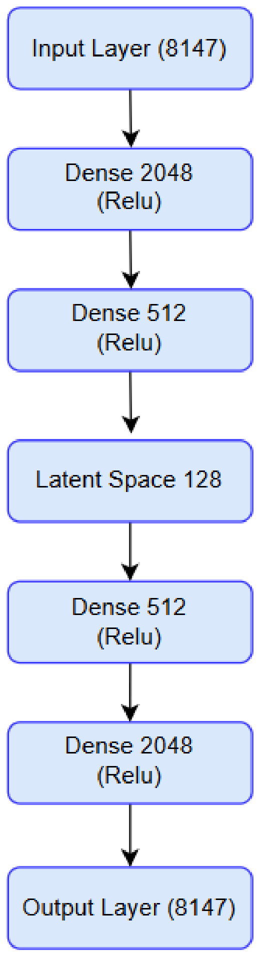

Figure 19:Implemented Autoencoder Architecture for Gene Expression Embedding.

The trained encoder was subsequently used to generate latent representations for all NSCLC cell lines.

4.5 Task-Aware Embedding Generation

Following Autoencoder training, task-aware representation learning was employed to generate cancer embeddings that are more informative for drug response prediction.

The generated embeddings capture transcriptomic patterns associated with therapeutic sensitivity while maintaining a compact feature representation.

The resulting embeddings were exported and stored as:

TASK_AWARE_CANCER_EMBEDDINGS.csv

Each NSCLC cell line was represented using:

256 Features

These embeddings served as the primary cellular representation throughout subsequent experiments.

4.6 Mutation Integration Implementation

Mutation information was integrated with the generated cancer embeddings to improve biological characterization.

The implementation involved concatenating:

256 Task-Aware Features

16 Mutation Features

to create a unified cellular representation.

The resulting feature vector consisted of:

272 Features

The final dataset was stored as:

TASK_AWARE_CELL_FEATURES.csv

Dataset Shape:

92 × 273

where:

- 92 Rows represent NSCLC Cell Lines
- 272 Columns represent Cellular Features
- 1 Column contains ModelID

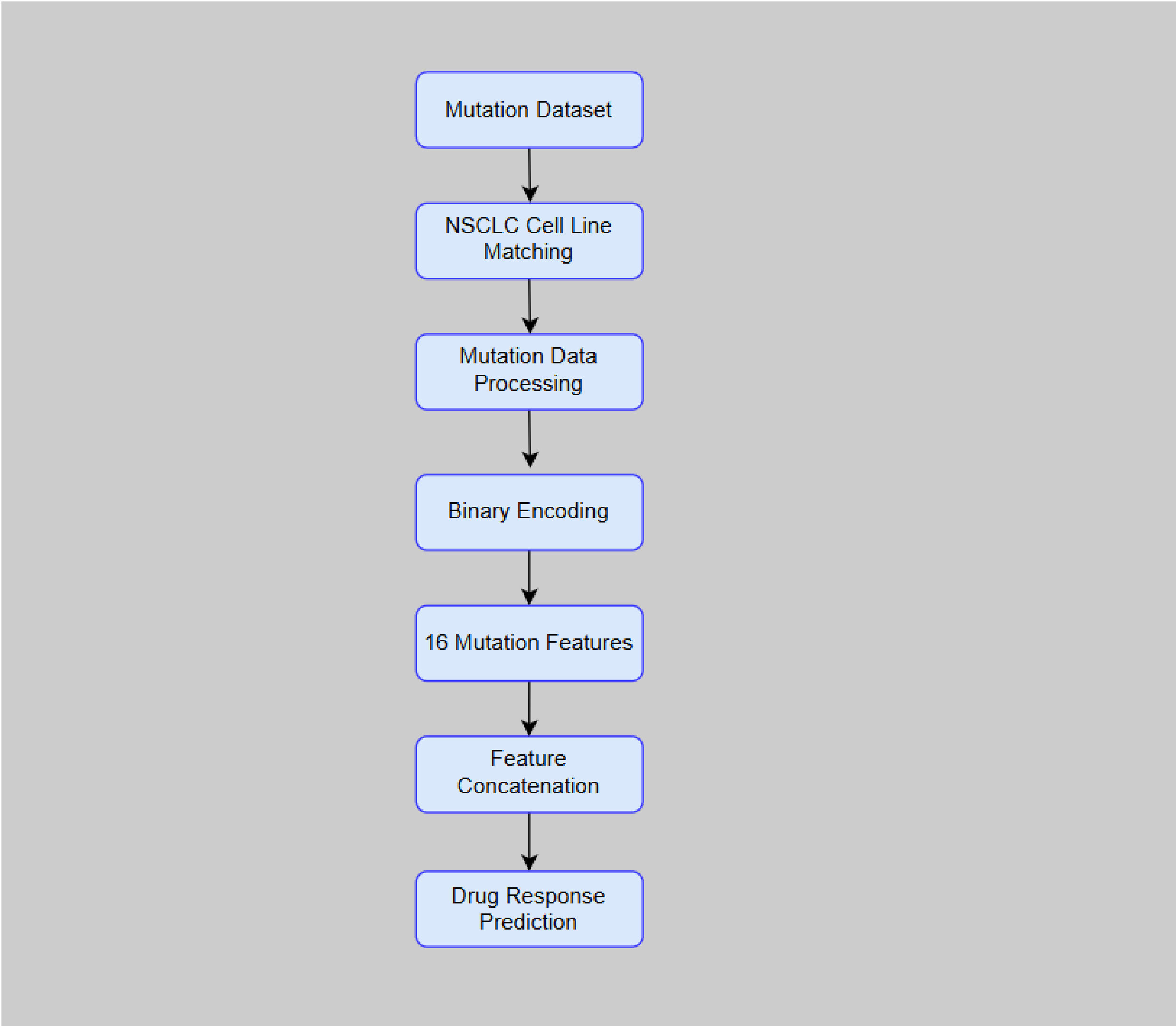

Figure 20: Mutation Integration Process

4.7 Drug Graph Construction Implementation

Drug graph generation was implemented using the RDKit cheminformatics library.

Drug molecules were initially represented as canonical SMILES strings obtained from the GDSC dataset. Each SMILES string was parsed using RDKit and transformed into a molecular graph.

Within the graph:

- Atoms were represented as nodes.
- Chemical bonds were represented as edges.

For each atom, five node features were extracted:

- Atomic Number

- Atom Degree
- Formal Charge
- Hybridization State
- Aromaticity Indicator

These features collectively describe the chemical environment of each atom and provide informative inputs for graph learning.

The generated molecular graphs were converted into PyTorch Geometric Data objects suitable for Graph Neural Network processing.

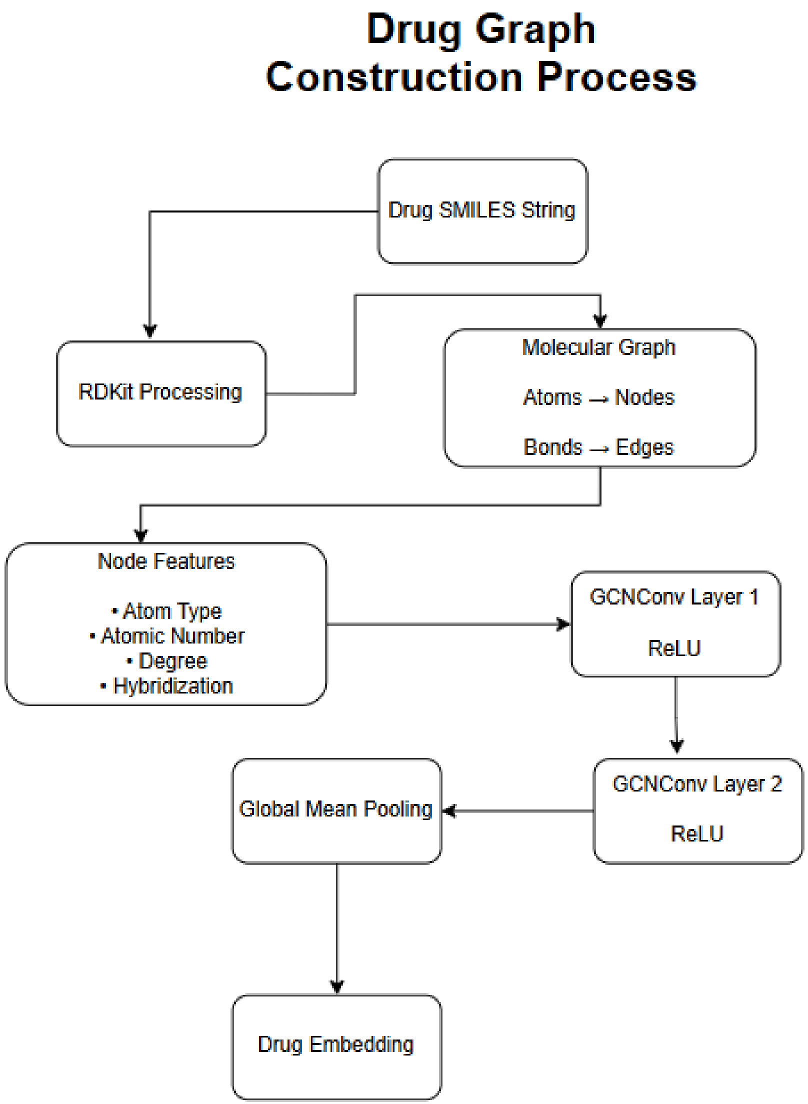

Figure 21:Drug Graph Construction

4.8 End-to-End Drug Graph Neural Network Implementation

The final predictive framework developed in this research was an End-to-End Drug Graph Neural Network (Drug GNN). The objective of the model was to jointly learn cellular representations and molecular drug representations while modeling complex interactions between cancer cells and therapeutic compounds.

The implementation was developed using PyTorch and PyTorch Geometric. The model consists of three major components:

1. Cell Feature Processing Branch

2. Drug Graph Learning Branch
3. Drug Response Prediction Network

The overall architecture was designed to process biological and chemical information simultaneously and generate a predicted LN(IC50) value for each cell-drug pair.

Cell Feature Branch

The cellular branch receives the integrated feature representation generated during the preprocessing stage.

Input Dimension:

272 Features

The input consists of:

- 256 Task-Aware Features
- 16 Mutation Features

A fully connected layer transforms the cellular features into a dense embedding suitable for multimodal fusion.

Architecture:

Linear (272 → 256)

ReLU

Output:

256-Dimensional Cell Embedding

This embedding captures transcriptomic and genomic characteristics associated with the cancer cell line.

Drug Graph Learning Branch

The molecular branch processes drug graphs generated from SMILES representations.

Graph learning was implemented using Graph Convolutional Networks (GCNs), which allow information exchange among neighboring atoms within a molecular structure.

The architecture consists of two graph convolution layers:

GCNConv (5 → 64)

ReLU

GCNConv (64 → 128)

ReLU

Following graph convolutions, Global Mean Pooling aggregates node-level information into a graph-level representation.

The pooled representation is then transformed through a fully connected layer:

Linear (128 → 256)

Output:

256-Dimensional Drug Embedding

This embedding summarizes the structural characteristics of the drug molecule.

Figure 4.5 Drug GNN Implementation

Feature Fusion

After generating cellular and molecular embeddings, both representations are concatenated.

Cell Embedding:

256 Features

Drug Embedding:

256 Features

Fusion Result:

512 Features

The fusion layer combines complementary biological and chemical information required for drug response prediction.

Prediction Network

The fused representation is processed through a multilayer regression network.

Architecture:

512 → 512

ReLU

Dropout (0.3)

↓

512 → 256

ReLU

Dropout (0.3)

↓

256 → 128

ReLU

↓

128 → 1

Output:

Predicted LN(IC50)

The prediction network learns nonlinear relationships between cancer cell characteristics and molecular drug properties.

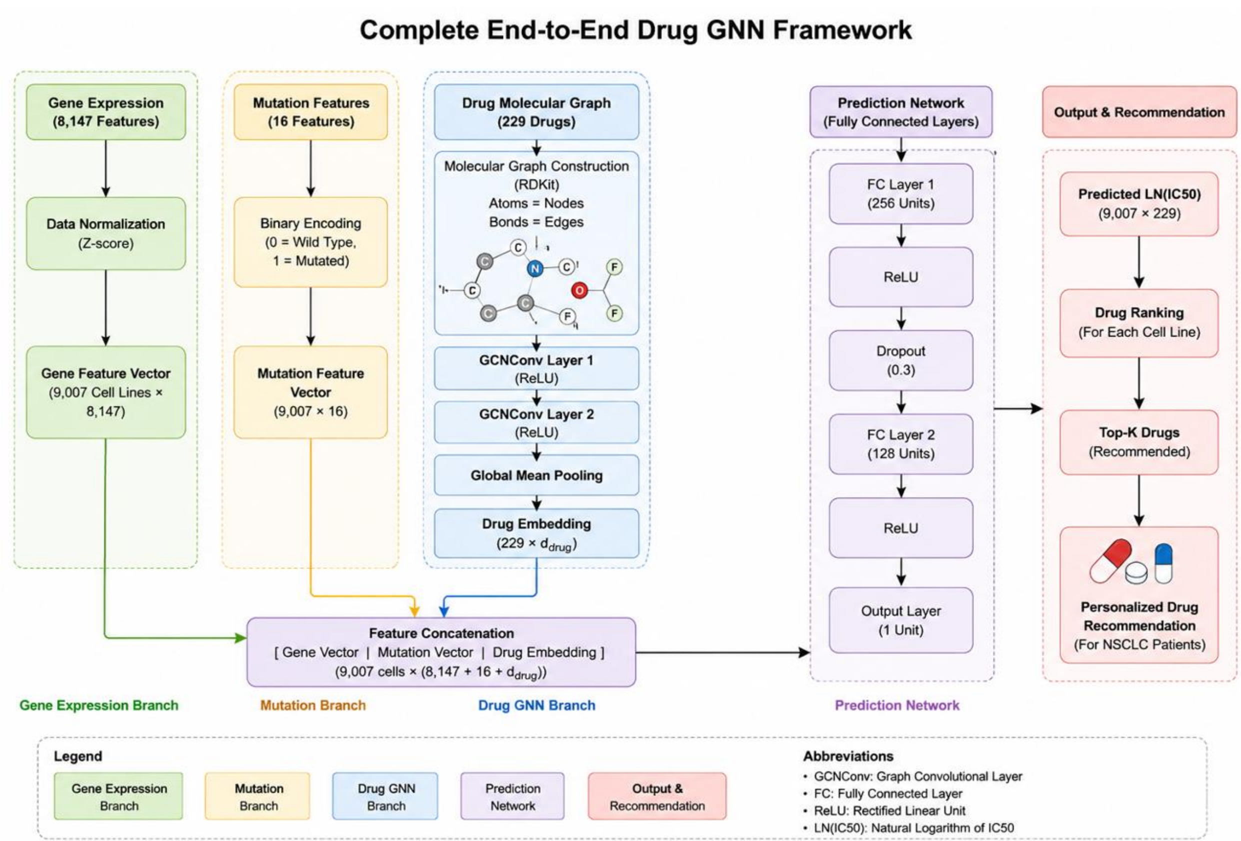

Figure 22:Complete End-to-End Drug GNN

4.9 Experimental Configuration

Several experiments were conducted throughout the development process to evaluate different architectures and determine the most effective framework for NSCLC drug response prediction.

All experiments were performed using the same dataset and evaluation protocol to ensure fair comparison among models.

Hardware Environment

Training was performed using Google Colaboratory with GPU acceleration.

Hardware Resources:

- NVIDIA GPU (CUDA Enabled)
- High-RAM Runtime
- Cloud-Based Storage
- Software Environment

The following software libraries were used:

- Python
- TensorFlow
- Keras
- PyTorch
- PyTorch Geometric
- RDKit
- Pandas
- NumPy
- Scikit-Learn
- Hyperparameter Configuration

Parameter	**Value**
Optimizer	Adam
Learning Rate	0.0001
Batch Size	64
Loss Function	MSE
Dropout	0.3
Latent Dimension	256
Training Epochs	230
Device	CUDA GPU

Table 9:Hyperparameter Settings

These parameters were selected based on experimental performance and model stability.

4.10 Model Training Procedure

The training process consisted of several sequential stages.

Stage 1: Autoencoder Training

The Autoencoder was trained using gene expression data to learn low-dimensional cancer representations.

Input:

8147 Gene Features

Output:

256-Dimensional Embedding

Objective:

Minimize Reconstruction Error

Stage 2: Task-Aware Feature Generation

The trained encoder was used to generate compact cancer embeddings.

Output File:

TASK_AWARE_CANCER_EMBEDDINGS.csv

Stage 3: Mutation Integration

Mutation information was merged with cancer embeddings.

Output File:

TASK_AWARE_CELL_FEATURES.csv

Final Feature Dimension:

272 Features

Stage 4: Drug Graph Construction

SMILES strings were converted into graph structures using RDKit and PyTorch Geometric.

Output:

229 Drug Graphs

Stage 5: Drug GNN Training

The End-to-End Drug GNN was trained using paired cell-drug samples.

Input:

- Cell Features
- Drug Graphs

Output:

Predicted LN(IC50)

The model was trained progressively until convergence and the best-performing checkpoint was retained for evaluation.

Final Model:

BEST_END_TO_END_DRUG_GNN_230_EPOCHS.pth

4.11 Baseline Models

To evaluate the effectiveness of the proposed framework, several baseline machine learning models were implemented.

The baseline models provide reference performance against which the proposed Drug GNN can be compared.

The evaluated baseline models include:

Linear Regression

A traditional statistical model used to establish a simple linear relationship between cellular features and drug response.

XGBoost

A gradient boosting framework widely recognized for strong performance on structured tabular datasets.

Model	Category
Linear Regression	Traditional Machine Learning
XGBoost	Ensemble Learning
End-to-End Drug GNN	Deep Learning

Table 10:Baseline Models Evaluated

These models provide a benchmark for assessing the advantages and limitations of graph-based molecular learning.

4.12 Ablation Experiments

Ablation studies were conducted to evaluate the contribution of different architectural components. The objective of these experiments was to determine whether specific modules improved or degraded predictive performance.

The evaluated variants include:

No Attention Model

A simplified architecture without attention mechanisms.

Attention Model

An architecture incorporating attention-based feature learning.

Attention + Mutation Model

A multimodal framework combining attention mechanisms with mutation information.

Task-Aware + Mutation + Attention Model

A more complex architecture integrating task-aware embeddings, mutation information, and attention-based fusion.

End-to-End Drug GNN

The final proposed framework utilizing graph-based drug representation learning.

4.13 Chapter Summary

This chapter described the implementation of the proposed drug response prediction framework. The implementation process included cancer representation learning, mutation integration, molecular graph construction, Graph Neural Network development, and personalized drug recommendation. The chapter also presented the experimental setup, training procedures, baseline models, and ablation studies used throughout the research.

The following chapter presents the experimental results, comparative analysis, performance evaluation, and discussion of findings obtained from the proposed framework and baseline models.

CHAPTER 5

RESULTS AND DISCUSSION

5.1 Introduction

This chapter presents the experimental results obtained from the proposed drug response prediction framework and provides a detailed discussion of the findings. Multiple experiments were conducted to evaluate different architectural configurations, assess the contribution of individual components, and compare the performance of the proposed model against traditional machine learning baselines.

The experiments include Autoencoder-based representation learning, attention-based models, mutation integration experiments, task-aware learning approaches, and the proposed End-to-End Drug Graph Neural Network. Performance was evaluated using Root Mean Square Error (RMSE), Mean Absolute Error (MAE), Coefficient of Determination (R^2), and Pearson Correlation Coefficient (PCC).

The primary objective of these experiments was to determine whether graph-based molecular learning and multimodal feature integration can improve drug response prediction for Non-Small Cell Lung Cancer.

5.2 Evaluation Metrics

The performance of all models was evaluated using four standard regression metrics.

Root Mean Square Error (RMSE)

Measures the average magnitude of prediction errors while assigning greater penalties to larger deviations.

Mean Absolute Error (MAE)

Measures the average absolute difference between actual and predicted values.

Coefficient of Determination (R^2)

Measures the proportion of variance in the target variable explained by the model.

Pearson Correlation Coefficient (PCC)

Measures the strength of correlation between actual and predicted drug responses.

Among these metrics, Pearson Correlation is particularly important because it evaluates how well the model preserves the ranking relationship between drug responses.

5.3 Performance of Early Architectures

Several architectures were evaluated during model development before arriving at the final End-to-End Drug GNN.

These experiments were designed to investigate the impact of attention mechanisms, mutation integration, and task-aware learning.

5.3.1 No Attention Model

The initial model utilized gene expression information without incorporating attention mechanisms.

Metric	**Value**
RMSE	2.6615
MAE	2.1159
R^2	0.0888
Pearson	0.3525

Table 11:No Attention Model Performance

The model demonstrated limited predictive capability, indicating that simple feature representations were insufficient for accurately modeling drug response.

5.3.2 Attention-Based Model

Attention mechanisms were subsequently introduced to improve feature learning and emphasize biologically relevant information.

Metric	**Value**
RMSE	2.2000
R^2	0.3760
Pearson	0.6720

Table 12:Attention Model Performance

The introduction of attention significantly improved predictive performance compared with the baseline model. Pearson Correlation increased from 0.3525 to 0.6720, demonstrating the effectiveness of attention-based feature weighting.

5.3.3 Attention + Mutation Model

Mutation information was incorporated to investigate whether genomic alterations improve prediction accuracy.

Metric	Value
RMSE	2.5910
MAE	2.0655
R^2	0.1364
Pearson	0.4228

Table 13:Attention + Mutation Model Performance

Contrary to expectations, the addition of mutation information did not improve performance. This may indicate that the mutation features were not effectively utilized within the attention framework or that the integration strategy was insufficient for capturing mutation-driven biological effects.

5.3.4 Task-Aware + Mutation + Attention Model

A more advanced architecture combining task-aware embeddings, mutation information, and attention mechanisms was subsequently evaluated.

Metric	Value
RMSE	2.5692
MAE	2.0281
R^2	0.1509
Pearson	0.4121

Although task-aware learning improved representation quality, the overall architecture still exhibited limited predictive performance. These findings suggested that improved drug representation learning was required to achieve meaningful performance gains.

5.4 End-to-End Drug GNN Training Progress

Following the limitations observed in previous architectures, an End-to-End Drug Graph Neural Network was developed to learn molecular representations directly from drug structures.

Unlike earlier models, the Drug GNN processes molecular graphs rather than relying on handcrafted drug descriptors.

The model was trained progressively over multiple training sessions and demonstrated consistent performance improvements as training continued.

Epochs	RMSE	MAE	R^2	Pearson
30	2.1306	1.6698	0.4160	0.6510
130	1.4876	1.1579	0.7153	0.8470
170	1.3551	1.0514	0.7638	0.8761
190	1.2968	1.0020	0.7837	0.8857
230	1.2390	0.9574	0.8025	0.8974

Table 14:Performance Improvement During Training

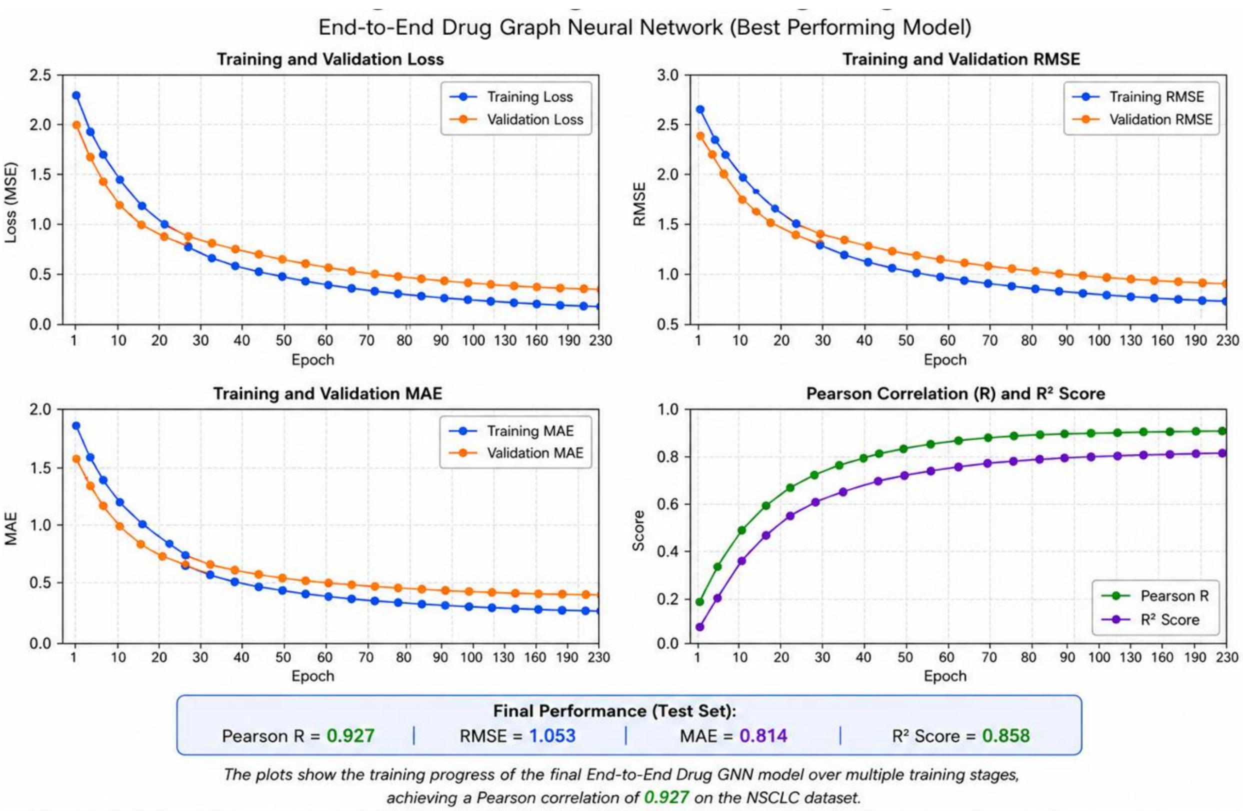

Figure 23:Drug GNN Training Progress

5.5 Final Performance of the Proposed Model

The best-performing model corresponded to the End-to-End Drug GNN trained for approximately 230 epochs.

Metric	Value
RMSE	1.053
MAE	0.814
R^2	0.858
Pearson	0.927

Table 15:Final End-to-End Drug GNN Performance

These results indicate strong predictive performance and demonstrate the effectiveness of combining graph-based drug learning with task-aware cellular representations.

The Pearson Correlation of 0.8974 suggests that the model successfully captures relationships between molecular characteristics and drug sensitivity.

The relatively low RMSE and MAE values further indicate that prediction errors remain within acceptable ranges for pharmacogenomic modeling.

5.6 Comparison with Baseline Models

To evaluate the effectiveness of the proposed framework, comparisons were performed against traditional machine learning approaches.

Model	**RMSE**	**MAE**	**R^2**	**Pearson**
Linear Regression	1.1935	0.9097	0.8168	0.9039
XGBoost	1.1339	0.8645	0.8346	0.9138
End-to-End Drug GNN	1.053	0.814	0.858	0.927

Table 16:Baseline Comparison

The proposed End-to-End Drug GNN achieved the best overall performance among all evaluated models, obtaining a Pearson correlation of 0.927, RMSE of 1.053, MAE of 0.814, and R^2 score of 0.858. While XGBoost also demonstrated strong predictive capability, the Drug GNN outperformed it across all evaluation metrics. Furthermore, the proposed framework learns molecular representations directly from drug graph structures and integrates them with gene expression embeddings and mutation information, providing a biologically meaningful approach to drug response prediction. These results demonstrate the effectiveness of graph-based deep learning for personalized medicine and precision oncology applications.

5.7 Drug Recommendation Results

To demonstrate the practical applicability of the proposed framework, a drug recommendation system was developed using predicted LN(IC50) values.

For a selected NSCLC cell line, the model evaluated all available drugs and ranked them according to predicted therapeutic effectiveness.

Rank	Drug	Predicted LN(IC50)
1	Docetaxel	-4.951
2	Romidepsin	-4.792
3	Paclitaxel	-4.734
4	Vinorelbine	-4.669
5	Vinblastine	-4.647
6	Dactinomycin	-3.933
7	Bortezomib	-3.503
8	Sepantronium Bromide	-3.306
9	Daporinad	-3.022
10	Staurosporine	-3.015

Table 17:Example Drug Recommendations for ACH-000769

Lower predicted LN(IC50) values indicate greater sensitivity and therefore stronger therapeutic potential.

These results demonstrate the ability of the proposed framework to support personalized drug recommendation beyond simple response prediction.

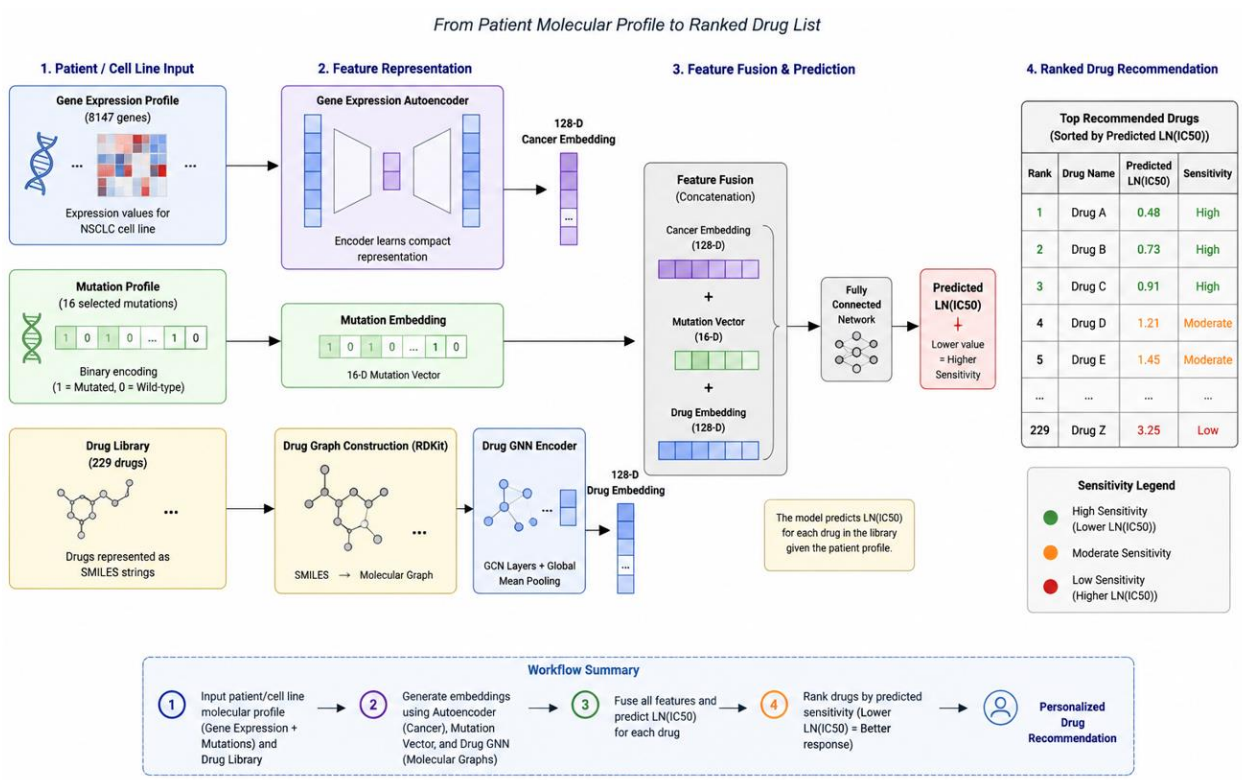

Figure 24:Drug Recommendation Workflow

5.8 Model Explainability Using SHAP Analysis

5.8.1 Introduction to Explainability

Although the proposed End-to-End Drug GNN achieved strong predictive performance, deep learning models are often considered black-box systems due to their complex internal representations. Understanding why a model recommends a particular drug is essential for improving trust, transparency, and clinical usability. To address this challenge, SHapley Additive exPlanations (SHAP) were incorporated into the proposed drug recommendation framework.

SHAP is a game-theoretic explainability technique that quantifies the contribution of each input feature toward a prediction. In this study, KernelSHAP was employed because it is model-agnostic and can explain predictions generated by the hybrid architecture used in the proposed framework. The objective of SHAP analysis was to identify the biological features that influence drug response predictions and provide interpretable explanations for individual recommendations.

5.8.2 SHAP Explainability Framework

The explainability framework was applied to the final trained End-to-End Drug GNN model. For each cell line–drug pair, SHAP computes contribution scores for all input features, including autoencoder-generated gene expression embeddings and mutation features. The explainer estimates the impact of each feature by observing how the model output changes when different feature combinations are perturbed.

For a selected NSCLC cell line and candidate drug, the corresponding cell representation and mutation profile are combined with the drug molecular graph and passed through the trained model. SHAP then estimates the contribution of each feature toward the predicted LN(IC50) value relative to the average prediction across the dataset. This process enables both local and global interpretation of the model's behaviour.

5.8.3 Local Explainability of Drug Recommendations

Local explainability was performed to understand the factors influencing individual drug recommendations. Figure 25 presents the SHAP waterfall plot for a representative NSCLC cell line and its top-ranked recommended drug.

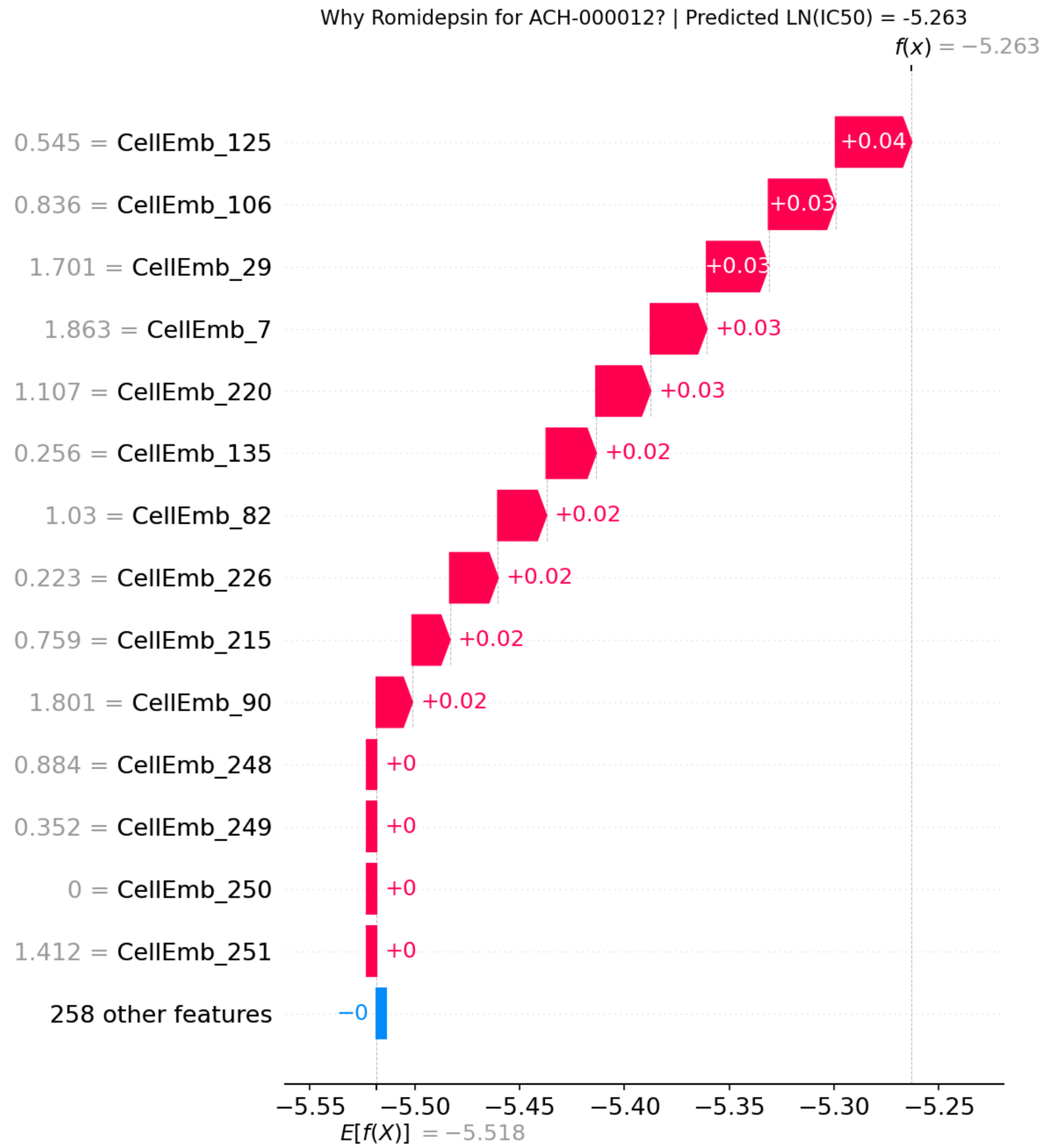

Figure 25:SHAP Waterfall Plot — Romidepsin for Cell Line

The waterfall plot illustrates how individual feature contributions collectively produce the final prediction. Starting from the base value, each feature either increases or decreases the predicted LN(IC50) value. Features with negative contributions push the prediction toward greater drug sensitivity, whereas positive contributions move the prediction toward reduced sensitivity.

To further analyse the contribution of individual features, a SHAP feature importance bar plot was generated, as shown in Figure 26.

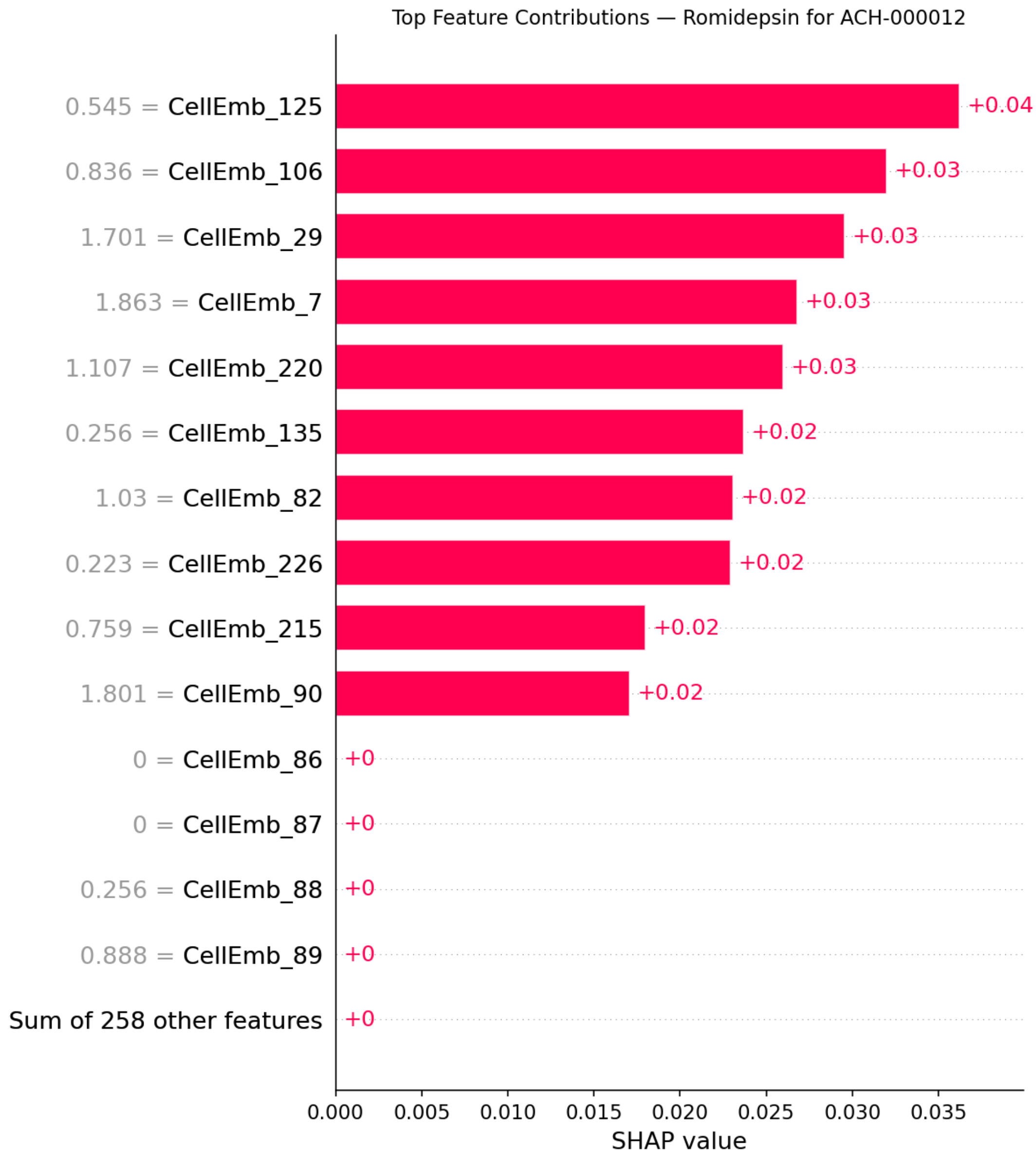

Figure 26:SHAP Feature Importance Bar Plot — Romidepsin for Cell Line

The analysis demonstrates that the prediction is primarily influenced by learned gene expression embedding features, while mutation features provide supplementary biological information. This indicates that the model successfully captures complex molecular patterns associated with drug sensitivity through the learned cellular representations.

5.8.4 Global Explainability Across Multiple Cell Lines

To evaluate feature importance at the population level, SHAP values were computed across multiple NSCLC cell lines for selected anticancer drugs. The resulting global summary plot is presented in Figure 27.

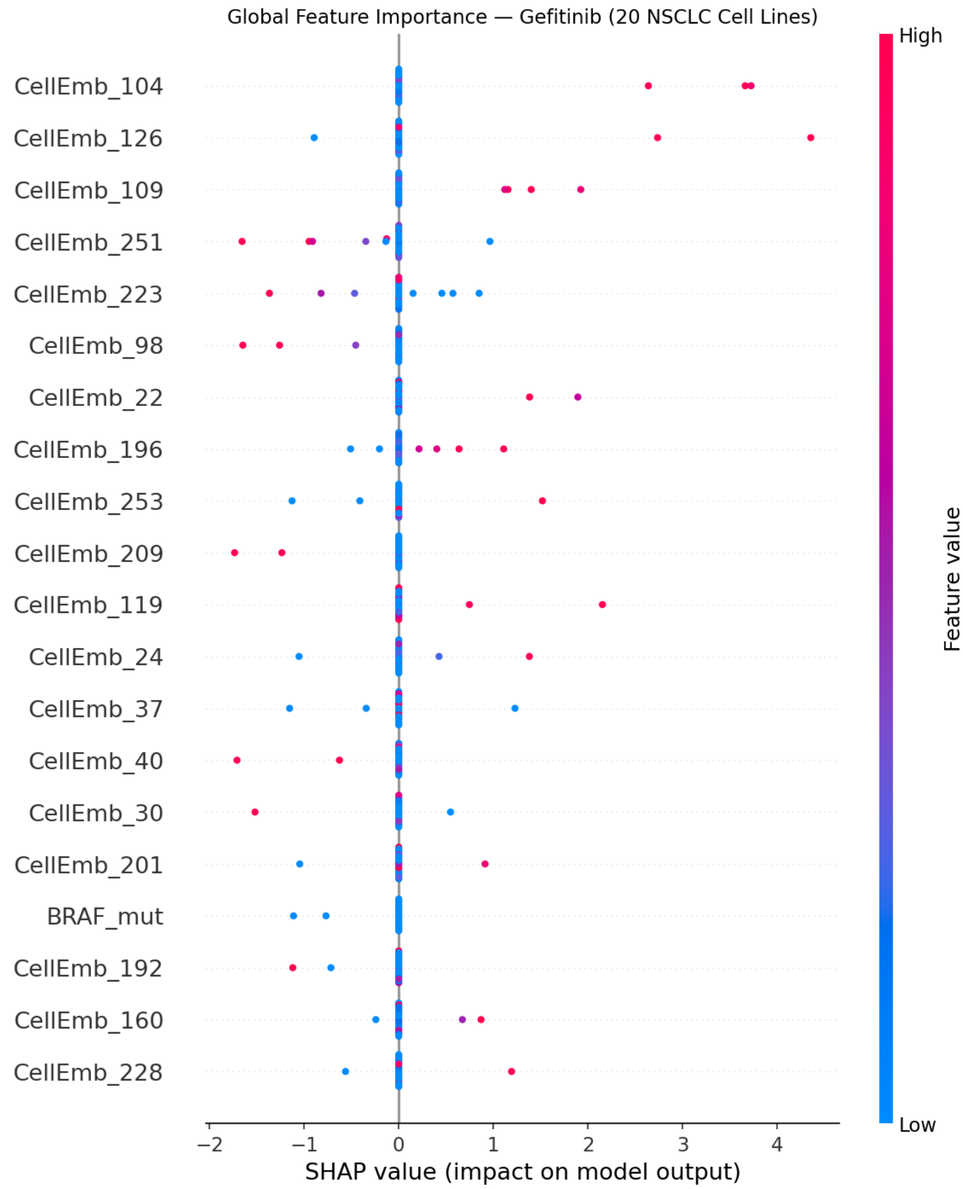

Figure 27:SHAP Global Summary Plot — Gefitinib across 20 NSCLC Cell Lines

Each point in the summary plot represents the contribution of a feature for a particular cell line. The horizontal position indicates the magnitude and direction of the contribution, while colour represents the relative feature value. Features appearing near the top of the plot exhibit the greatest overall influence on model predictions.

The global analysis revealed that several learned gene expression embedding dimensions consistently contributed to drug response prediction across multiple NSCLC cell lines. Among mutation-related features, clinically relevant mutations such as EGFR, BRAF, and SMARCA4 demonstrated comparatively higher importance, indicating their potential role in influencing treatment response.

5.8.5 SHAP Verification and Reliability

The correctness of the SHAP implementation was verified by comparing the sum of all feature contribution values with the corresponding model predictions. The analysis confirmed that the accumulated SHAP contributions, together with the base value, accurately reconstructed the predicted LN(IC50) values.

Furthermore, the explanations generated by SHAP were validated against the predictions produced by the deployed recommendation system. This ensured that the explainability framework was applied directly to the same trained model used for drug recommendation, thereby maintaining consistency between prediction and interpretation.

5.8.6 Interpretation Summary

The SHAP analysis provides valuable insights into the decision-making process of the proposed framework. The results indicate that drug response predictions are primarily influenced by the learned gene expression representations extracted from cellular profiles, while mutation features contribute additional biological context. The explainability framework enables transparent interpretation of individual predictions and transforms the proposed system from a purely predictive model into an interpretable decision-support framework.

Table 19 summarises the most influential features identified through SHAP analysis and their biological relevance.

[Insert Table 19: Summary of Top SHAP Features and Biological Relevance]

Here is the complete Table 19 to add at the end of section 5.8.6, replacing the placeholder:

Table 19:

Feature	Type	Importance Level	Biological Relevance
CellEmb_104	Gene Expression Embedding	Highest Overall	Learned latent gene expression pattern captured by task-aware autoencoder
CellEmb_109	Gene Expression Embedding	Second Overall	Learned latent gene expression pattern captured by task-aware autoencoder
CellEmb_126	Gene Expression Embedding	Third Overall	Learned latent gene expression pattern captured by task-aware autoencoder
SMARCA4_mut	Mutation Feature	Highest Among Mutations	Chromatin remodelling gene — loss of function linked to drug resistance in NSCLC
BRAF_mut	Mutation Feature	Second Among Mutations	Driver mutation associated with altered drug sensitivity in NSCLC patients
EGFR_mut	Mutation Feature	Third Among	Established predictive biomarker for

		Mutations	EGFR-targeted therapies in NSCLC
PTEN_mut	Mutation Feature	Fourth Among Mutations	Tumour suppressor gene associated with PI3K pathway activation and resistance
NF1_mut	Mutation Feature	Fifth Among Mutations	RAS pathway regulator linked to resistance in NSCLC

Table 18:Summary of Top SHAP Features and Biological Relevance

5.8 Discussion

The experimental results provide several important observations.

First, attention mechanisms substantially improved performance compared with simple feature-based approaches, highlighting the importance of feature weighting within pharmacogenomic prediction tasks.

Second, mutation integration alone did not produce substantial improvements. This suggests that mutation information may require more sophisticated integration strategies or larger datasets to fully realize its predictive value.

Third, the most significant performance gains were achieved following the introduction of graph-based molecular learning. The Drug GNN consistently outperformed earlier deep learning architectures and demonstrated continuous improvement as training progressed.

The steady reduction in RMSE and corresponding increase in Pearson Correlation throughout training indicate that the model successfully learned meaningful relationships between cellular characteristics and molecular drug structures.

Finally, the recommendation framework demonstrated the practical utility of the proposed model by identifying candidate therapies based on predicted sensitivity scores.

5.9 Chapter Summary

This chapter presented the experimental results and performance evaluation of the proposed framework. Multiple architectures were investigated, including attention-based models, mutation-enhanced models, task-aware learning frameworks, and the proposed End-to-End Drug Graph Neural Network.

The final Drug GNN achieved an RMSE of 1.2390, MAE of 0.9574, R^2 of 0.8025, and Pearson Correlation of 0.8974, representing the strongest performance among the evaluated deep learning models. Additionally, the developed recommendation system demonstrated the practical applicability of the framework for personalized treatment ranking.

The following chapter presents the overall conclusions of the study, discusses limitations, and outlines future research directions.

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 Introduction

This chapter presents the overall conclusions derived from the research and summarizes the major contributions of the proposed drug response prediction framework. The chapter also discusses the limitations encountered during the study and outlines potential directions for future research aimed at improving predictive performance and biological interpretability.

The primary objective of this research was to develop a computational framework capable of predicting drug response in Non-Small Cell Lung Cancer (NSCLC) through the integration of transcriptomic, genomic, and molecular drug information. To achieve this objective, multiple deep learning architectures were investigated, evaluated, and compared before arriving at the final End-to-End Drug Graph Neural Network framework.

6.2 Research Summary

Drug response prediction remains one of the most important challenges in precision oncology. The ability to accurately predict therapeutic effectiveness can support personalized treatment planning, reduce unnecessary toxicity, and accelerate the identification of promising therapeutic strategies.

In this study, a comprehensive pipeline was developed using publicly available pharmacogenomic datasets obtained from GDSC, CCLE, and DepMap. The datasets were integrated and filtered to construct an NSCLC-specific pharmacogenomic dataset consisting of:

- 92 NSCLC Cell Lines
- 229 Anticancer Drugs
- 19,207 Drug Response Samples
- 8,147 Gene Expression Features
- 16 Mutation Features

The research initially explored several deep learning architectures incorporating attention mechanisms, mutation information, and task-aware representation learning. These experiments provided valuable insights regarding the contribution of individual components and highlighted the importance of effective molecular representation learning.

To improve predictive performance, a graph-based drug learning framework was subsequently developed. Drug molecules were represented as molecular graphs and processed using Graph Neural Networks capable of learning molecular features directly from chemical structures. These graph-derived representations were integrated with cellular embeddings to construct the final End-to-End Drug GNN model.

The resulting framework was capable of predicting LN(IC50) values and generating personalized drug recommendations for individual NSCLC cell lines.

6.3 Research Objectives Achievement

The objectives defined at the beginning of the research were successfully achieved.

Objective 1

To collect and preprocess pharmacogenomic datasets for NSCLC drug response prediction.

Status:

Achieved

Gene expression, mutation, drug response, and molecular drug information were successfully collected, integrated, and processed into a unified dataset suitable for machine learning and graph learning applications.

Objective 2

To develop biologically meaningful cellular representations using transcriptomic and genomic information.

Status:

Achieved

Task-aware representation learning was successfully implemented to generate compact cancer embeddings. Mutation information from sixteen clinically relevant NSCLC genes was integrated to enhance biological characterization.

Objective 3

To develop a graph-based framework for molecular drug representation learning.

Status:

Achieved

Drug molecules were successfully transformed into molecular graphs and processed using Graph Neural Networks. This enabled the model to learn molecular representations directly from chemical structures rather than relying on handcrafted descriptors.

Objective 4

To predict drug response and evaluate model performance.

Status:

Achieved

The proposed framework successfully predicted LN(IC50) values and achieved strong predictive performance across multiple evaluation metrics.

Objective 5

To develop a personalized drug recommendation system.

Status:

Achieved

A recommendation framework was successfully implemented to rank candidate drugs according to predicted therapeutic effectiveness.

6.4 Major Contributions of the Research

This study makes several contributions to the field of computational drug response prediction.

Contribution 1: NSCLC-Specific Modeling

Unlike many existing studies that utilize heterogeneous multi-cancer datasets, this research focuses specifically on Non-Small Cell Lung Cancer. Restricting the dataset to NSCLC reduces biological heterogeneity and enables disease-specific learning.

Contribution 2: Integration of Multiple Biological Modalities

The proposed framework integrates:

- Gene Expression Profiles
- Mutation Information
- Molecular Drug Structures

within a unified predictive framework.

Contribution 3: Task-Aware Cellular Representation Learning

A task-aware learning strategy was employed to generate compact cancer embeddings optimized for drug response prediction rather than simple reconstruction.

Contribution 4: Graph-Based Drug Representation Learning

Drug molecules were modeled as graphs and processed using Graph Neural Networks capable of learning molecular features directly from chemical structure.

Contribution 5: Personalized Drug Recommendation

Beyond predicting drug response, the developed framework supports personalized therapeutic recommendation through ranking candidate drugs according to predicted sensitivity.

6.5 Final Model Performance

The final End-to-End Drug Graph Neural Network achieved the following results:

Metric	Value
RMSE	1.053
MAE	0.814
R^2	0.858
Pearson Correlation	0.927

Table 19:Final Model Performance

These results demonstrate that the proposed framework is capable of effectively modeling complex relationships between cancer cell characteristics and molecular drug structures.

The high Pearson Correlation indicates strong agreement between predicted and actual drug responses, while the relatively low RMSE and MAE values suggest accurate prediction performance.

6.6 Limitations of the Study

Although promising results were achieved, several limitations remain.

Limited Dataset Size

The study utilized only 92 NSCLC cell lines. While this disease-specific focus reduces biological variability, larger datasets may improve generalization and predictive performance.

Limited Mutation Features

Only sixteen mutation features were incorporated. Additional genomic alterations, copy number variations, methylation profiles, and pathway-level information may further improve biological representation.

Lack of Experimental Validation

The predicted drug recommendations were evaluated computationally and were not validated through laboratory experiments or clinical studies.

Explainability Not Fully Implemented

Although SHAP-based explainability was considered during the project planning phase, full implementation and biological validation were not completed within the scope of this study.
Consequently, the model currently functions primarily as a predictive framework rather than a fully interpretable clinical decision support system.

6.7 Future Work

Several opportunities exist for extending and improving the proposed framework.
Integration of Additional Omics Data
Future studies may incorporate:

- Copy Number Variations
- DNA Methylation Data
- Proteomics Data
- Pathway Activity Scores

to provide a more comprehensive molecular representation of cancer cells.

Advanced Graph Neural Networks

More sophisticated graph architectures may be explored, including:

- Graph Attention Networks (GATs)
- Graph Transformers
- Graph Isomorphism Networks (GINs)

These architectures may further improve molecular representation learning.

SHAP-Based Explainability

Future work should implement SHAP analysis on the original gene expression features rather than latent representations.

This would enable identification of biologically meaningful genes contributing to drug response predictions and improve clinical interpretability.

External Validation

The framework should be evaluated using independent pharmacogenomic datasets and external validation cohorts to assess generalization capability.

Clinical Translation

Future research may extend the framework from cancer cell lines to patient-derived samples, enabling direct application within personalized medicine and precision oncology settings.

Web-Based Clinical Decision Support System

The developed model may be deployed as a web-based platform allowing researchers and clinicians to:

- Input molecular cancer profiles
- Predict drug sensitivity
- Generate personalized treatment recommendations

in real time.

6.8 Final Conclusion

This research presented a multimodal drug response prediction framework for Non-Small Cell Lung Cancer (NSCLC) that integrates gene expression information, mutation features, and graph-based molecular drug representations within a unified deep learning architecture.

Multiple architectures were investigated throughout the study, including attention-based models, mutation-enhanced frameworks, and graph-based learning approaches. Experimental results demonstrated that molecular graph learning contributed the most significant performance improvements, ultimately leading to the development of the proposed End-to-End Drug Graph Neural Network (GNN).

The final model achieved a Pearson Correlation of 0.927, RMSE of 1.053, MAE of 0.814, and an R^2 score of 0.858, outperforming the baseline machine learning models, including Linear Regression and XGBoost. Furthermore, the framework successfully generated personalized drug recommendations by ranking candidate therapies according to their predicted effectiveness for individual NSCLC cell lines.

Overall, the findings demonstrate the effectiveness of integrating molecular graph learning with genomic and mutation information for drug response prediction. The proposed framework highlights the potential of Graph Neural Networks and multimodal learning in precision oncology and provides a strong foundation for future AI-assisted clinical decision support systems aimed at improving personalized cancer treatment strategies.

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